Genetec™ Security Center.

Sipelia™ User Guide 2.14

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Genetec™

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You can send your comments, corrections, and suggestions about this guide to documentation@genetec.com.

About this guide

This guide is intended for Sipelia™ administrators. It describes how to set up, configure, and manage the Sipelia™ module as part of your Security Center system.

Notes and notices

The following notes and notices might appear in this guide:

- **Tip:** Suggests how to apply the information in a topic or step.
- **Note:** Explains a special case or expands on an important point.
- **Important:** Points out critical information concerning a topic or step.
- **Caution:** Indicates that an action or step can cause loss of data, security problems, or performance issues.
- **Warning:** Indicates that an action or step can result in physical harm, or cause damage to hardware.

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1

Getting started

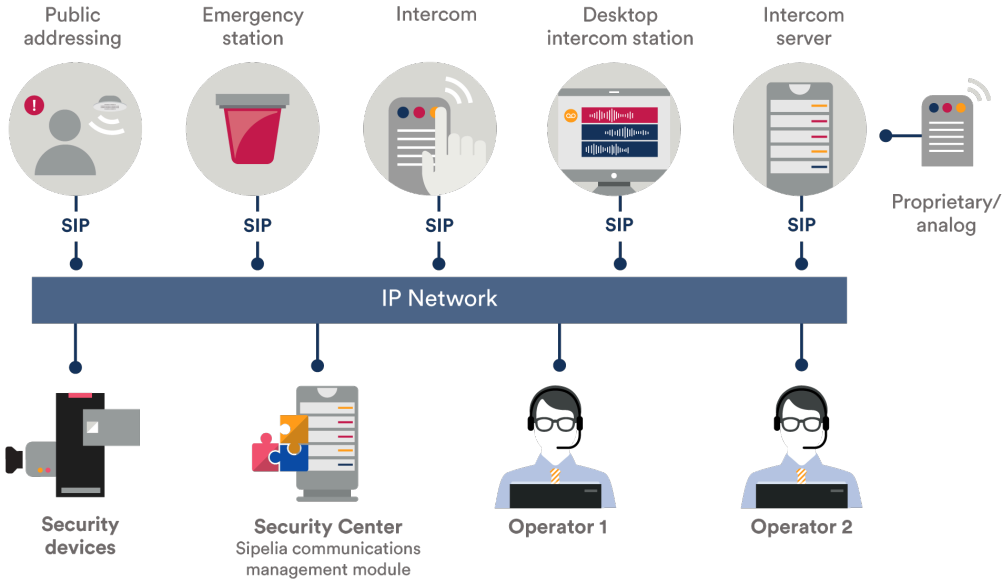
This section includes the following topics:

- "What is Sipelia?" on page 2
- "How licensing works in Sipelia" on page 4
- "Deploying Sipelia " on page 5
- "User interface tour of Sipelia Client" on page 6
- "Common VoIP terms" on page 8

What is Sipelia?

Sipelia™ is a core module of Security Center. Using this module, Security Center users can make, receive, and manage SIP-based voice and video calls over a network. Running on the open source Session Initiation Protocol (SIP), Sipelia also integrates existing video and access control platforms with intercom systems, and allows users to log call activities.

Overview of Sipelia communications management



Main features

With Sipelia™ installed in Security Center, you can do the following:

- Make voice and video calls through Security Center.
- Connect standard USB headsets and webcams to Security Desk workstations.
- Receive incoming call notifications directly through the notification tray in Security Desk.
- Initiate, answer, forward, place on hold, or cancel calls from a dedicated call dialog box.
- Generate reports to investigate the activities within specific call sessions.
- Generate *Unified reports* to search and combine events from different Sipelia entities.
- Create dashboards to visualize *call session* and intercom status information.
- Watch call sessions that have associated video.
- Control cameras, doors, zones, and device outputs during a call.
- Deploy a SIP-based solution that makes it easy to use your existing communications infrastructure.
- Connect to SIP intercom devices, intercom exchange servers, and mobile apps through the SIP standard.
- Create a customizable list of contacts, so that users can quickly call their contacts. Contact lists can include other Security Center users, ring groups, workstations, and SIP devices.
- Create ring groups so that multiple Security Center users and *SIP entities* can receive incoming calls at the same time or one after another until a user takes the call.
- Enable authentication for incoming calls.
- Play live, pre-recorded, or text messages in emergency situations using public addressing.

- Receive and make calls using the Genetec™ Mobile and Genetec™ Web App applications.

Typical applications

A Sipelia integration within Security Center can help you in the following applications:

- Respond to and investigate an emergency.
- Respond to employees who have lost their access cards.
- Grant access to high-security rooms.
- Monitor and manage who enters and exits a parking lot.
- Make communication more efficient with user-to-user video calls.
- Broadcast messages in various situations.

How licensing works in Sipelia

Sipelia™ requires a set of licenses to enable the installation, use, and advanced features of the plugin in Security Center.

Security Center 5.11 and later requires the following license:

- **GSC-Sipelia-1SIP:** The license is required to install the Sipelia plugin on Security Center. It allows you to make and receive calls, record call sessions, add a SIP trunk, and broadcast messages to any SIP entity managed by Sipelia™. You need one license per SIP device that you are adding to your system.

Security Center 5.10 and earlier requires the following licenses:

- **GSC-Sipelia-Base:** The base system license is required to install the Sipelia plugin on Security Center and to allow the operators to make and receive calls.
- **GSC-Sipelia-1SIP-STD:** The standard license allows a SIP device (such as a SIP intercom) to be used in Security Center, whether it registers directly to Sipelia Server or is made available via a SIP trunk. You need one standard license per SIP device that you are adding to your system.
- **GSC-Sipelia-1SIP-ADV:** The advanced license enables the recording of call sessions and failover of the Sipelia plugin role to a second server when needed. You need one advanced license for each standard license that you are adding to your system.

  NOTE: If the standard and advanced license counts do not match, the system chooses the lower count for both types of licenses. See the example below.
- **GSC-Sipelia-1Trunk:** The trunk license allows you to add and configure a SIP trunk. You need one trunk license for each SIP trunk that you are adding to your system.
- **GSC-Sipelia-1PA:** The Public Address functionality allows you to broadcast a live or pre-recorded message to any SIP entity managed by Sipelia. You can use a maximum of one public address license per system.

Example

The following sample license for Sipelia 5.10 and earlier shows how the system applies the license counts when they do not match:

License type	Number of licenses installed	Number of licenses applied
GSC-Sipelia-Base	1	1
GSC-Sipelia-1SIP-STD	100	50
GSC-Sipelia-1SIP-ADV	50	50
GSC-Sipelia-1Trunk	4	4
GSC-Sipelia-1PA	1	1

Deploying Sipelia

To integrate SIP-based communication into Security Center, so that users can communicate through VoIP, you can deploy Sipelia™ as part of your Security Center system.

Before you begin

- Read the Sipelia™ 2.14 Release Notes.
- Familiarize yourself with the ports used by Sipelia modules in Security Center.
- Familiarize yourself with the common VoIP terms and the Sipelia glossary of terms that are used throughout this guide.

Procedure

1 Install Sipelia Server.

2 Create the Sipelia plugin role.

3 Define the ranges of your SIP phone extensions.

4 Configure the audio and video recording of call sessions.

5 Configure SIP accounts for your Security Center users.

6 Create user ring groups.

7 Add your SIP intercoms.

8 Register your SIP intercoms with Sipelia Server.

9 Create ring groups.

10 Install Sipelia Client on each of the Security Desk workstations that run Sipelia.

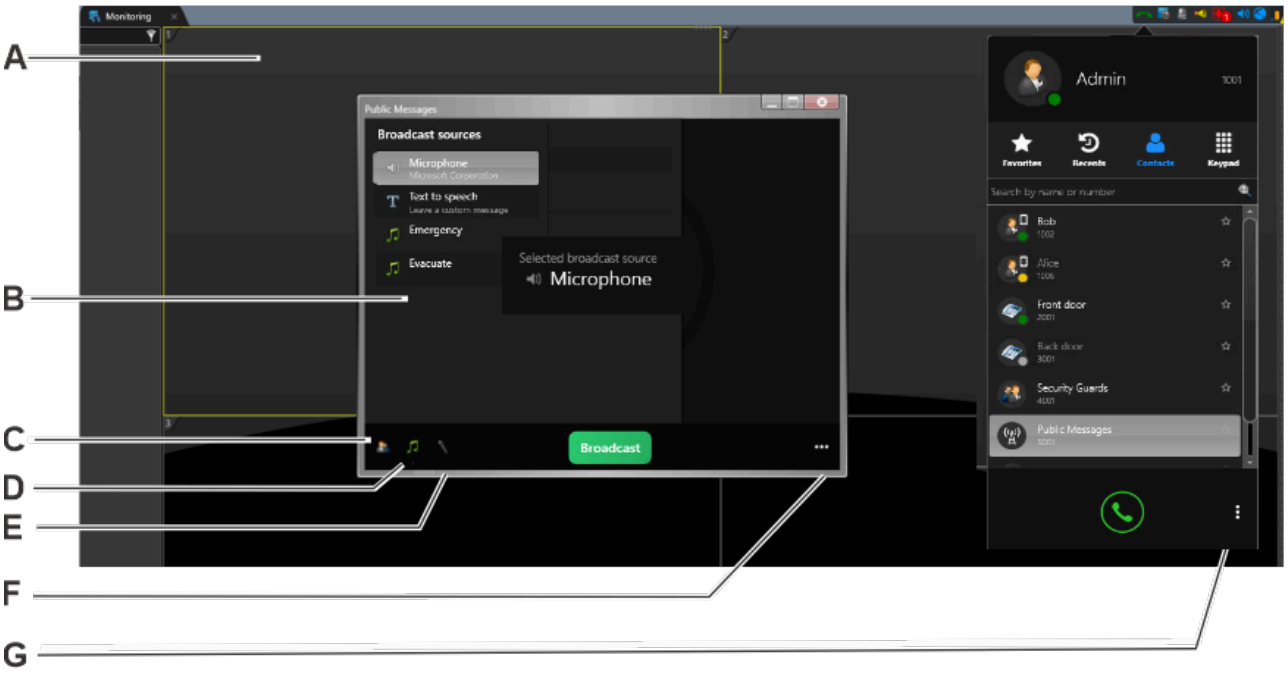
11 Configure your devices for voice and video calls in Security Desk.

12 Configure two-way communication between Sipelia Server and other SIP servers.

User interface tour of Sipelia Client

Sipelia™ is a core module of Security Center. Using this module, Security Center users can make, receive, and manage SIP-based voice and video calls over a network.

Sipelia™Client must be installed on every Security Desk workstation that runs Sipelia software, thus turning Security Desk into a SIP client, or softphone. It installs the various user interface features of the Sipelia module, such as the call dialog box and conversation window. The following image shows some of the features with an example of a public address call.



A	**Canvas**	View and control entities in tile mode or window mode.
B	**Conversation window**	Manage different types of calls and view associated video when available.
C	**Contact list**	Contact lists can include other Security Center users, ring groups, workstations, and SIP devices.
D	**Audio**	Select the audio to broadcast in a public address.
E	**Microphone**	Enable or disable the microphone.
F	**Change view**	Switch to change to a tile view.
G	**Call dialog box**	Manage calls, list of contacts, favorites, and availability status.

Example

Watch this video to learn more. Click the **Captions** icon (**CC**) to turn on video captions in one of the available languages.



Call management in Sipelia

Integrating Sipelia™ within your Security Center environment provides you with an array of features to help you manage your calls directly from Security Desk.

Watch this video to learn more. Click the **Captions** icon (**CC**) to turn on video captions in one of the available languages.



Call report task in Sipelia

In the *Call report* task, users can investigate call sessions, view the call logs for all sessions, watch playback video and audio of call sessions, and switch between the different video sources that are linked to a session.

Watch this video to learn more. Click the **Captions** icon (**CC**) to turn on video captions in one of the available languages.



Common VoIP terms

A VoIP (voice over internet Protocol) environment needs many interacting components to work properly. Familiarizing yourself with these common terms can help you gain a better understanding of the Sipelia™ module.

- **Dual-tone multifrequency (DTMF):** Dual-tone multifrequency is the standard for the audible signal that is sent to the phone company once a key is pressed on a telephone keypad. There's an audible tone to represent each digit on a keypad.
- **IP phone:** An internet protocol (IP) phone is a device that is used to make and receive calls over the internet. An IP phone can use any of the existing communication standards or protocols, such as SIP, to transmit calls across a network. Although an IP phone can look like a traditional phone, an IP phone isn't connected to a phone-line jack found in typical POTS installations; an IP phone is connected to a router or RJ-45 Ethernet connector.
- **Plain old telephone system (POTS):** The plain old telephone system (POTS) is the basic form of telephone service that is used by most homes and businesses worldwide. Apart from being a different technology, what separates a non-POTS service from a POTS service is speed and bandwidth. POTS is also known as the public switched telephone network (PSTN).
- **Private branch exchange (PBX):** A private branch exchange (PBX) is a private phone network that is used within a company. A PBX is a switching station that is used to connect many internal phone extensions to one outside line, thereby making it more efficient and cost effective for companies to adopt a phone system. In a company that uses a PBX network, incoming calls are redirected by the PBX to one or more internal extensions within the same enterprise.
- **IP private branch exchange (IP PBX):** An IP PBX is a private branch exchange (telephone switching system within an enterprise) that switches calls between VoIP (voice over internet Protocol or IP) users on local lines while allowing all users to share some external phone lines.
- **SIP trunking:** SIP trunking is the process of using VoIP technology to connect existing PBX systems to other PBX systems. SIP trunking replaces traditional phone trunks with an IP network and consolidates voice, data, and video into a single trunk (or line). SIP trunking ensures a more reliable communication service and cost reductions.
- **Softphone:** A softphone is a software for managing inbound and outbound calls over a network, using your computer rather than a phone. Softphones are designed to simulate the functions found on traditional phones. Also known as *SIP client*.
- **Voice over Internet Protocol (VoIP):** Voice over Internet Protocol (VoIP) is the technology for routing two-way voice and video communications over the web or other IP networks.
- **Traversal Using Relays around NAT (TURN):** The Traversal Using Relays around NAT (TURN) server enables communication between the Sipelia Gateway role and mobile devices that are behind *network address translators (NATs)* or firewalls.

2

Installation

This section includes the following topics:

- "About Sipelia components" on page 10
- "Supported codecs" on page 12
- "Installing Sipelia Server " on page 13
- "Uninstalling Sipelia Server" on page 15
- "Installing Sipelia Client" on page 16
- "Uninstalling Sipelia Client" on page 17
- "Automating Sipelia installation" on page 18

About Sipelia components

The Sipelia™ module has three components: the Server, the Client, and Sipelia Gateway. Sipelia Server integrates SIP functions into Security Center and allows your system to store data. Sipelia™ Client turns Security Desk into a SIP client and enables the various features. Sipelia Gateway allows you to access Security Desk features from your mobile device or web browser.

Sipelia Server

Sipelia™ Server is the SIP server component of Sipelia. It receives and administers information about the different SIP endpoints, and essentially facilitates the communication between two or more endpoints that are communicating in a SIP environment. Sipelia Server also collates and stores important data, such as contact list information, SIP server settings, and call session recordings. *Sipelia™ Server* is a server plugin () that must be run by a Security Center plugin role. It must be installed on the Security Center server where you intend to host this plugin role.

Sipelia Server stores the following data:

- User options
- List of contacts
- Session information (such as users, time and duration of phone calls, and associated entities)
- Audio and video files related to call sessions
- Configurations of SIP phone extensions for users and devices
- SIP server settings
- Ring group configuration
- SIP trunk configuration
- Dial plan configuration
- Recording settings for users and devices
- Security Center events that are linked to the SIP intercom entity
- Paging zone configuration

Sipelia Client

Sipelia Client is the softphone component of Sipelia. It installs the various user interface features of the Sipelia module.

Sipelia Client installs the following features:

- Notification tray
- Call dialog box
- Conversation window
- Call report task

Sipelia Gateway

The Sipelia™ Gateway component is a separate plugin that depends on the Sipelia Server and offers Sipelia functionality inside Genetec™ Mobile and Genetec™ Web App.

You can access the following features through your mobile device or web browser:

- Receive and make calls over a private network or cellular data.
- Access your contact list and favorites.
- Receive incoming call notifications.

- Use standard call functions such as forward, hold, and mute.

Supported codecs

Sipelia™ supports several codecs for audio and video communication.

SIP intercoms must support at least one audio codec and one video codec from the following list. In cases where both sides of the communication use a different codec, Sipelia Server transcodes the media stream to ensure that communication is possible.

Supported video codecs:

- H.264
- H.263
- H.263+ (1998)

Supported audio codecs:

- PCMA (G.711a)
- PCMU (G.711u)
- telephone-event
- iLBC
- GSM
- Speex (Narrowband)
- Speex (Wideband)
- Speex (Ultrawideband)
- L.16
- L.16-44-1
- G.722
- G.723
- G.726-16
- G.726-24
- G.726-32
- G.726-40
- G.728
- G.729
- Opus

NOTE: PCMA (G.711a), PCMU (G.711u), and G722 are enabled by default.

Supported video resolutions:

- Video resolutions up to 720p
- Resolutions 1080p or higher:
  - Add the intercom as a camera in Security Center and attach it to the intercom.
  - Turn off the SIP video on the device and the default video is available from the Security Center camera. Configure the SIP video to a lower resolution and video streams are available from the call window.

Related Topics

High-quality audio codecs on page 52

Installing Sipelia Server

To integrate SIP functions into Security Center and allow your system to store data such as phone book contacts and the recordings of call sessions, you must first install Sipelia™ Server and configure the module in Config Tool.

Before you begin

- Your servers must meet the hardware requirements. For more information, see System requirements for Sipelia on the TechDoc Hub.
- Config Tool must be installed on the system on which you plan to install Sipelia Server.
- It's recommended to install Sipelia Server on a dedicated Security Center expansion server. For information about how to add an expansion server to your Security Center system, see Adding expansion servers on the TechDoc Hub.
- If you already have Sipelia installed (2.0 or earlier) and you're upgrading to a newer version (2.1 or later) and you're also upgrading to a new version of Security Center (from 5.3 or earlier to 5.4 or later), delete the Sipelia plugin role before upgrading Security Center. You can then install your new version of Sipelia and make it point to your previous database.

What you should know

- Sipelia Server is installed by default with Security Center 5.12.0.0 and later. The following procedure only applies to Security Center versions 5.11.3.0 and earlier.
- Genetec™ Update Service (GUS) is supported in Sipelia version 2.11 and later. GUS updates Sipelia when a new release becomes available.
- *Sipelia™ Server* is a server plugin ( ) that must be run by a Security Center plugin role. It must be installed on the Security Center server where you intend to host this plugin role.
- You can have a maximum of one Sipelia role per Directory.
- Network interfaces used by the plugin to communicate with Config Tool and Security Desk must be enabled in Server Admin.

Procedure

1 Download the product from GTAP.

You need a username and password to log on to GTAP.

2 Double-click **setup.exe**.

The product's *InstallShield Wizard* dialog box opens.

3 Select the installation language and click **Next**.

This language selection does not limit the language availability of the installed software. The Sipelia user interface appears in the language that is selected for Security Center.

4 Click **Next**.

5 Read the license agreement, accept the terms, and click **Next**.

6 Select a folder location to install the product, and click **Next**.

7 In the *Custom Setup* dialog box, select the components you want to install.

- Sipelia™ Client installations require the *Client* component.
- Sipelia Server installations require the *Server* component.
- Sipelia Gateway installations require the *Gateway Plugin Services* components.

8 Click **Next**.

The installation can take a few minutes.

9 After completion, select **Restart Genetec Server** and click **Finish**.

IMPORTANT: Restart the *Genetec™ Server* for the system to detect that a new plugin is installed. Restarting the Genetec™ Server causes a short interruption of the Security Center service on the server. If you can't afford to interrupt the service now, you can restart the Genetec Server later, as long as you do so before configuring Sipelia in Config Tool. To avoid interruptions, it's recommended to install Sipelia Server on a dedicated Security Center expansion server.

Example

Watch this video to learn more. Click the **Captions** icon (**CC**) to turn on video captions in one of the available languages.

After you finish

In Config Tool, create the Sipelia plugin role.

Uninstalling Sipelia Server

To completely remove Sipelia™ Server from your system, you must perform a series of steps.

Procedure

1 Close all Security Center applications (Security Desk, Config Tool, and Server Admin).

2 Click **Start** > **Control Panel** > **Programs and Features**.

3 In the *Programs and Features* window, right-click **Genetec Sipelia Installer**, and then click **Uninstall**.

NOTE: There are two applications associated with Sipelia: *Genetec Sipelia Installer*, and *Genetec Sipelia X.X*. Uninstalling *Genetec Sipelia Installer* removes both applications.

4 In the *Remove the Program* dialog box, click **Remove**.

5 When the message *Uninstallation Completed* is displayed, click **Finish**.

6 Restart the computer.

The Genetec™ Sipelia Server application is removed from the system.

Installing Sipelia Client

To turn Security Desk into a SIP client and use the various features of the Sipelia™ module, you must install Sipelia™ Client on every Security Desk workstation that runs Sipelia.

Before you begin

- Install Sipelia Server on your Security Center system.
- Install Security Center Client on the computer that you want to install Sipelia Client on.

What you should know

- Sipelia Client is installed by default with Security Center 5.12.0.0 and later. The following procedure only applies to Security Center versions 5.11.3.0 and earlier.
- Genetec™ Update Service (GUS) is supported in Sipelia version 2.11 and later. GUS updates Sipelia when a new release becomes available.

Procedure

1 Download the product from GTAP.

You need a username and password to log on to GTAP.

2 Double-click **setup.exe**.

The product's *InstallShield Wizard* dialog box opens.

3 Select the installation language and click **Next**.

This language selection does not limit the language availability of the installed software. The Sipelia user interface appears in the language that is selected for Security Center.

4 Click **Next**.

5 Read the license agreement, accept the terms, and click **Next**.

6 Select a folder location to install the product, and click **Next**.

7 In the *Custom Setup* dialog box, expand the **Client** node.

8 If your system uses the legacy Plan Manager 10.3 plugin, select **Plan Manager Intercom Object** if you want to add SIP intercoms to the map.

Do not select this option if your system uses the new *Maps* task that is integrated into Security Center.

9 Click **Next**.

The installation can take a few minutes.

10 When completed, click **Finish**.

11 Restart Security Desk and Config Tool.

Example

Watch this video to learn more. Click the **Captions** icon (**CC**) to turn on video captions in one of the available languages.

After you finish

If deploying Sipelia, configure your audio and video devices, so that Security Center users can make and receive voice and video calls.

Uninstalling Sipelia Client

To completely remove Sipelia™ Client from your system, you must perform a series of steps.

Procedure

1 Close all Security Center applications (Security Desk, Config Tool, and Server Admin).

2 Click **Start** > **Control Panel** > **Programs and Features**.

3 In the *Programs and Features* window, right-click **Genetec Sipelia Installer**, and then click **Uninstall**.

NOTE: There are two applications associated with Sipelia: *Genetec Sipelia Installer*, and *Genetec Sipelia X.X*. Uninstalling *Genetec Sipelia Installer* removes both applications.

4 In the *Remove the Program* dialog box, click **Remove**.

5 When the message *Uninstallation Completed* is displayed, click **Finish**.

Automating Sipelia installation

You can perform a silent installation of the Sipelia™ product (Client and Server) by running an installer command with proper arguments.

What you should know

A silent installation is an automated way of installing software without user intervention. The silent installation is run from the command prompt using the *setup.exe* executable, and Windows Installer commands. You can specify which Sipelia installation parts should be deployed by using the command arguments.

The following table lists the command-line arguments.

Argument	Description
`/silent`	Informs the installer that the installation should be silent.
`/ISFeatureInstall`	Specifies which parts of Sipelia should be deployed. The following values are possible: • Sipelia: obligatory part if the Sipelia™ Client or Server is required for deployment. • Client: optional, instructs the installer to deploy the Sipelia Client. • Server: optional, instructs the installer to deploy the Sipelia Server. • Gateway and Gateway_Services: optional, instructs the installer to deploy the Sipelia Gateway Plugin Services component.
`/debuglog"C:\log.txt"`	Optional: Creates a log file from the silent installation process under the following location: C:\log.txt.

Procedure

1 In Windows, right-click the **Command Prompt**, and then click **Run as administrator**.

2 Navigate to the folder where the SipeliaInstaller.exe executable is located. By default, it is in the 'Sipelia Packages' folder.

3 In the Command Prompt, run the following command:

Example:

```
"SipeliaInstaller.exe" /silent /DebugLog"C:\Temp\Logs\Sipelia.log" /log"C:\Temp
\Logs\" /ISFeatureInstall=Sipelia,Client,Server,Gateway,Gateway_Services.txt
```

NOTE: This command installs Sipelia Server, Client, and Gateway. The installation log is stored in C:\SipeliaSilentInstallationLog.txt.

4 SipeliaInstaller.exe" /silent /DebugLog"C:\Temp\Logs\Sipelia.log" /log"C:\Temp\Logs\" /ISFeatureInstall=Sipelia,Client,Server,Gateway,Gateway_Services

3

Configuration

This section includes the following topics:

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Creating the Sipelia plugin role

After you have installed Sipelia™ Server on a Security Center server machine, you can create the Sipelia plugin role in Config Tool.

Before you begin

Make sure that Sipelia Server is installed.

Procedure

1 From the Config Tool homepage, open the *Plugins* task.

2 Click Add an entity (), and select **Plugin**.

The *Creating a role: Plugin* wizard appears.

3 In the **Server** drop-down, select the server that will host the Sipelia Server role.

4 In the **Select plugin type** field, select **Sipelia**™ or **Sipelia**™ **Gateway** and add each role separately.

5 Enter the values for **Database server** and **Database** for the Sipelia database or use the default values and click **Next**.

The default values are already provided.

6 Enter the entity name and description, and then select the desired partition for this role and click **Next**.

7 Check that the information that you've entered is correct.

8 Click **Create**, and then click **Close**.

Sipelia appears in the list of plugin roles (). It can take a few minutes for the role to create its database.

Example

Watch this video to learn more. Click the **Captions** icon (**CC**) to turn on video captions in one of the available languages.

Defining the ranges of SIP phone extensions

Before assigning a SIP extension for a given user, ring group, or SIP intercom, you can define multiple ranges of SIP extensions, and then choose a different range for each SIP entity.

What you should know

The phone extension ranges are sets of SIP extensions from which you can assign an extension number to each of your SIP entities. Each extension range must have a default password for the extensions, you can only have a maximum of 1,000 extensions per range, and you must have a minimum of one defined extension range to be able to connect a SIP entity to Sipelia™. By default, Sipelia provides five extension ranges (*Range 1* to *Range 5*), each with a default password of 1234.

Procedure

1 From the Config Tool homepage, open the *Plugins* task.

2 Select the Sipelia plugin role, and click **Server**.

3 Note the extension ranges that are already defined, and decide on how to assign them to your various SIP entities.

4 To add an extension range, click **Add range**.

5 Enter the following:

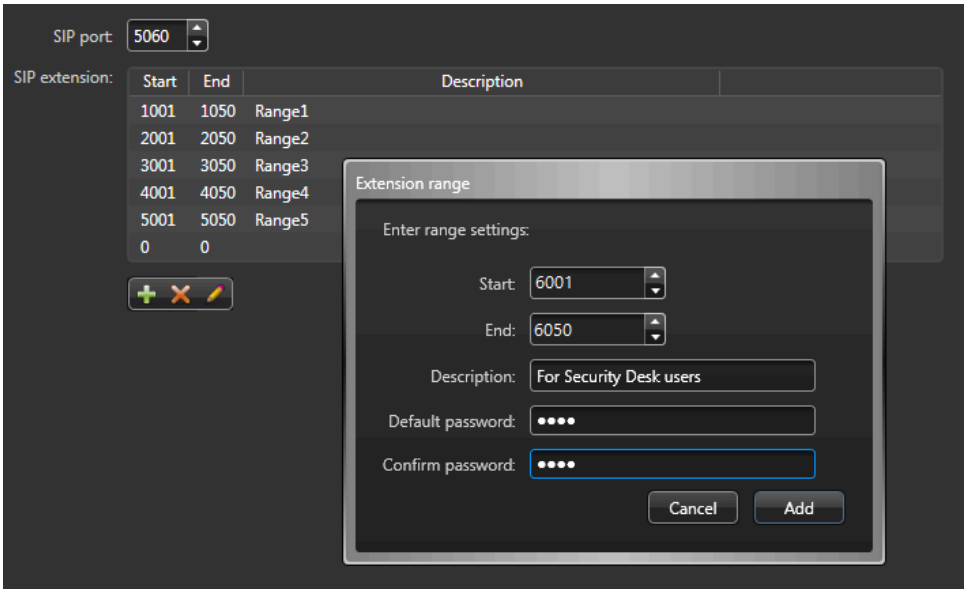
- **Start:** The start value of the SIP extension range. The start value must be unique and can't be greater than the end value.
- **End:** The end value of the SIP extension range. The end value must be unique and can't be less than the start value.
- **Description:** A phrase that describes the range, and perhaps indicates what SIP entity the range is reserved for.
- **Default password:** The password for every SIP extension within the range. All SIP entities whose respective extensions lie within this range must know this password. Every SIP endpoint, such as softphones and SIP intercoms, that needs to connect to the Sipelia™ Server must have this password value (along with the extension) in their respective configurations.
- **Confirm password:** The confirmation of the default password. Values in both password fields must match.

NOTE: The start and end values for the extension ranges are inclusive.

6 Click **Add**, and then click **Apply**.

Example

As shown in the following image, let's assume that you want to define an extension range reserved solely for Security Center users, and you want to limit the number of extensions to 50. Add a unique extension range that spans 50 numbers, and then enter a default password that applies to every extension within this range. When configuring SIP accounts for your Security Center users, you can assign each user an extension number within this new range, but you can only assign a maximum of 50 users to this range.



Watch this video to learn more. Click the **Captions** icon (**CC**) to turn on video captions in one of the available languages.

After you finish

If deploying Sipelia, configure the recording of the audio and video of call sessions.

Configuring TLS

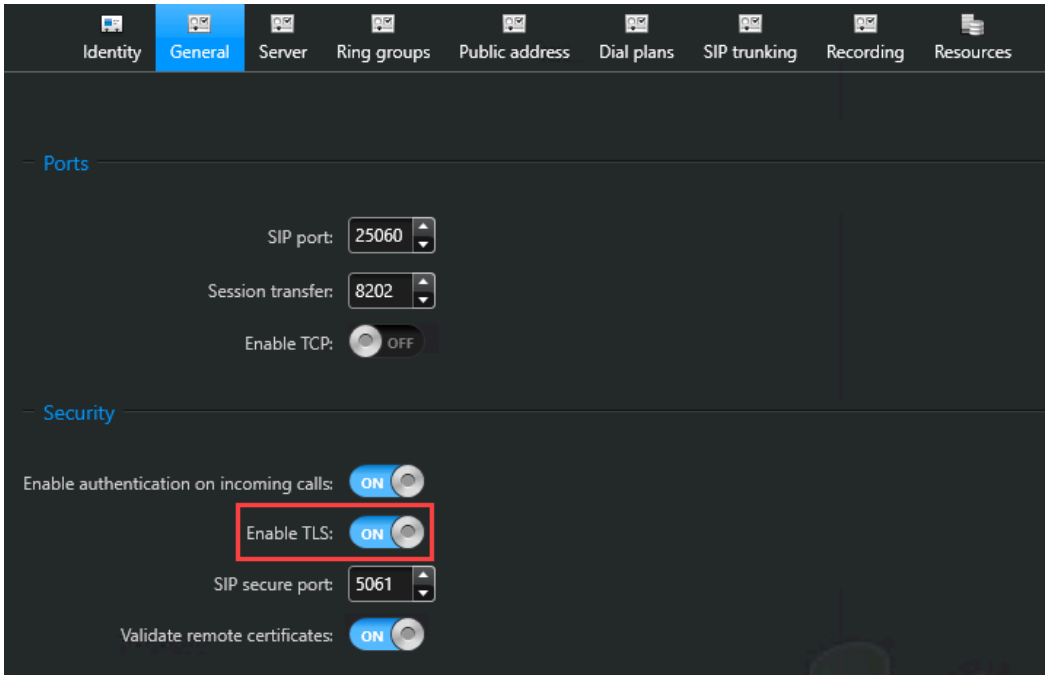
To secure the communication over a network, you must configure Sipelia™ to accept TLS connections.

Procedure

1 From the Config Tool homepage, open the *Plugins* task.

2 Select the Sipelia plugin and click the **General** tab.

3 Turn on the **Enable TLS** option.
This option is turned off by default.



4 Enter the SIP secure port number.
The default is 5061.

5 (Optional) If you want to use *self-signed certificates*, disable the **Validate remote certificate** option.

6 Click **Apply**.

Configuring SIP accounts for Security Center users

Security Center users can communicate with each other using the SIP-related controls in Security Desk. To enable this, you must configure a SIP account for each of your users and assign the appropriate privileges.

Before you begin

- Create the users that you want to configure SIP accounts for. For more information, see Creating users.
- Ensure that all users have access rights to the partition where the Sipelia™ plugin is located. For more information, see Granting access rights for partitions.
- If you don't want to use the default extension ranges that are already provided, define your own ranges of SIP phone extensions.

What you should know

After configuring a SIP account for a Security Center user, that user becomes a SIP entity. A SIP entity is a Security Center entity that has SIP-related capabilities. In Security Center, examples of SIP entities are users, ring groups, and SIP devices such as SIP intercoms.

Procedure

1 From the Config Tool homepage, open the *User Management* task.

2 Select a user from the list.

3 Click the **VoIP** tab to set up this SIP entity as a SIP endpoint.

4 Assign a SIP extension to your SIP entity in one of the following ways:

- Click **Auto-assign**. Auto-assign automatically assigns the SIP entity to the next available phone extension in a given range. After you click this button, choose an existing range and click **Apply**.

  NOTE: Auto-assign is the recommended way of assigning a SIP extension. It automatically populates the correct password for the range.
- Enter a SIP extension manually:
  - **SIP extension:** The SIP entity's phone extension. To be able to communicate with other SIP endpoints, every SIP entity (user, ring group, or intercom) in Security Center must have a unique SIP extension assigned to it.
  - **Password:** The password for the extension. This password was specified when creating the extension range. Each SIP extension within a given range has its password automatically set to match the default password for that range.

  IMPORTANT: If you want to change the extension password, you must first change it in the intercom device portal, and then apply the same password to the SIP intercom in the *Area view* .

5 Assign a call priority:

- **Critical:** When an operator receives a call with this priority, a red bar is displayed beside their name in the call list. When multiple calls are received simultaneously, calls with this priority are displayed at the top of the call list.
- **Urgent:** When an operator receives a call with this priority, a yellow bar is displayed beside their name in the call list. When multiple calls are received simultaneously, calls with this priority are displayed below critical calls in the call list.
- **Normal:** When an operator receives a call with this priority, there's no color associated with the user in the call list. When multiple calls are received simultaneously, calls with this priority are displayed below critical calls in the call list.

NOTE: Security Center Automations can be configured based on the call priority of an incoming call. Also, you can filter reports for SIP calls by call priority in the *Unified Report* and *VoIP Inventory* tasks.

6 Configure the following settings:

- **Record audio and video:** Allows you to record the call sessions that the SIP entity participates in (either caller or recipient of a call). Once recorded, sessions can later be reviewed and exported through the Call report task. The default value is inherited from the global recording settings which are found on the *Recording* page of the Sipelia Plugin role. Changing this setting at the entity level forces the entity to no longer inherit the value from the global setting, thus allowing you to turn recording on or off for only specific entities, without affecting all of them.
- **Roaming profile:** When enabled (default value), it stores a user's respective Security Desk option settings in the database. Users can sign in to Security Desk from different workstations on the same network and still keep their personalized settings. For example, the option to have incoming calls always open in a tile.
- **Gateway enabled:** When enabled, users can access Sipelia features in Genetec™ Mobile and Genetec™ Web App.

7 Click the *Privileges* tab to set up the user's privileges for Sipelia.

8 Under **All privileges** > **Application privileges** > **Sipelia**, select the privileges corresponding to the actions that the user may perform.

IMPORTANT: By default, privileges are set to *Undefined*. For users to make and receive calls, you must explicitly grant the appropriate privileges.

9 Click **Apply**.

Example

Watch this video to learn more. Click the **Captions** icon (**CC**) to turn on video captions in one of the available languages.

After you finish

If deploying Sipelia, create user ring groups.

Allowing users to see pictures of other users

If you want Security Center users to see the picture of other users in Sipelia™, you must manually add them in the security settings of the associated custom field.

Before you begin

- Create the Sipelia plugin role.

What you should know

When created, the Sipelia plugin role automatically adds the *Photo* custom field in Security Center. The picture of each user, shown in the list of contacts for example, is provided by this custom field. To be able to see the pictures, users must be added to the *Security* property of the custom field.

Procedure

1 From the Config Tool homepage, open the *System* task.

2 Click **Custom fields**, select *Photo* in the list, and then click **Edit the item** ( ).

3 Under *Security*, click **Add an item** ( ), and then select the user.

BEST PRACTICE: You can add all your Sipelia users to a single user group, and then add this group to the custom field. This way, each time new users are added to the group, the pictures will automatically be made visible to them.

4 Click **OK** > **Save and close** > **Apply**.

NOTE: You can create more custom fields that include information such as location, title, and team. This information is available by clicking in the call dialog box.

Associating Security Center cameras with users

To extend your monitoring capabilities within Security Desk, you can associate Security Center cameras users that have SIP accounts, so that the live video stream from Security Center cameras is displayed during active phone calls between these users.

Before you begin

- Create your Security Center users (see the *Security Center Administrator Guide* for details).
- Configure SIP accounts for your Security Center users.

What you should know

Video streams from a Security Center camera do not require a SIP connection, and are therefore not considered video calls. To configure video calls between SIP entities, you must configure the required audio and video devices in Security Desk.

As opposed to SIP intercoms, which allow you to associate cameras, doors, zones, and output devices to them, you can only associate a camera to a Security Center user.

Procedure

1 From the Config Tool, the open*User Management* task.

2 Select a user from the list.

3 Click the **VoIP** tab.

4 In the **Associated entities** section, click **Add an entity** ( ).

5 In the *Entity association wizard* search for and select the camera entity that you want to associate to your user, and then click **Next**. If you select a PTZ camera, you can select the PTZ preset whose video stream you want to appear during an active call.



6 Confirm your selection by clicking **Next**, and then **Apply**.

7 Once the *Entity association wizard* has closed, click **Apply** to save your changes.
The entity that you selected appears in the list of associated entities.

8 (Optional) To configure the associated entity that you added, click .

9 (Optional) Associate custom events to call states or Sipelia entity states.

10 Associate additional cameras to your user by repeating the steps above.

The camera you have selected is linked to the user. During an active call, the live video stream from this camera appears in the conversation window or Security Desk tile of the users with which this user is communicating. If you have associated multiple cameras, users are able to switch between the different video streams.

Example

Let's assume that you have associated the following two cameras to user Charles: *Front Building Entrance* and *Emergency Exit*. As shown in the following image, when Charles is in an active call with Security Center users, those users can switch between the live video streams of both cameras.


Front Building Entrance (1st)
Emergency Exit (1st) - 02

Adding SIP intercoms

To increase the security of your premises and grant access to people upon confirming their identities, you can add a SIP intercom and associate it to entities, such as a camera or door.

Before you begin

- Install the SIP intercom according to the recommendations of the intercom manufacturer.
- If you want to connect a SIP intercom to a *SIP trunk* and not a Sipelia™ Server add a SIP trunk, and then define associated dial plan rules.
- Define a range of phone extensions for your SIP intercoms. It's recommended to dedicate extension ranges for each particular SIP entity, especially intercoms that communicate with Sipelia Server through a SIP trunk.

What you should know

A SIP intercom is an intelligent SIP endpoint that provides two-way phone connectivity in a SIP environment. In Security Center, a SIP intercom is one of the established SIP entities, and it is the only SIP entity that is an actual device. The other SIP entities are Security Center users and ring groups.

Procedure

1 From the Config Tool homepage, open the *Plugins* task.

2 Select the **Sipelia**™ plugin role.

3 At the bottom of the page, click **Add intercom** ( ).

4 Enter a descriptive name for your SIP intercom, and then click **Add**.
The *Area view* task opens, and the intercom you added appears in the list of entities.

5 Click the **VoIP** tab to set up this SIP entity as a SIP endpoint.

6 Assign a SIP extension to your SIP entity in one of the following ways:

- Click **Auto-assign**. Auto-assign automatically assigns the SIP entity to the next available phone extension in a given range. After you click this button, choose an existing range and click **Apply**.
  NOTE: Auto-assign is the recommended way of assigning a SIP extension. It automatically populates the correct password for the range.
- Enter a SIP extension manually:
  - **SIP extension:** The SIP entity's phone extension. To be able to communicate with other SIP endpoints, every SIP entity (user, ring group, or intercom) in Security Center must have a unique SIP extension assigned to it.
  - **Password:** The password for the extension. This password was specified when creating the extension range. Each SIP extension within a given range has its password automatically set to match the default password for that range.

  IMPORTANT: If you want to change the extension password, you must first change it in the intercom device portal, and then apply the same password to the SIP intercom in the *Area view* .

7 Assign a call priority:

- **Critical:** When an operator receives calls with this priority, a red bar is displayed beside the device's name in the call list. When multiple calls are received simultaneously, calls with this priority are displayed at the top of the call list.
- **Urgent:** When an operator receives calls with this priority, a yellow bar is displayed beside the device's name in the call list. When multiple calls are received simultaneously, calls with this priority are displayed below critical calls in the call list.
- **Normal:** When an operator receives calls with this priority, there's no color associated with the user. When multiple calls are received simultaneously, calls with this priority are displayed below critical calls in the call list.

NOTE: Security Center Automations can be configured based on the call priority of an incoming call. Also, you can filter reports for SIP calls by call priority in the *Unified Report* and *VoIP Inventory* tasks.

8 Set the following:

- **Record audio and video:** Allows you to record the call sessions that the SIP entity participates in (either caller or recipient of a call). Once recorded, sessions can later be reviewed and exported through the Call report task. The default value is inherited from the global recording settings which are found on the *Recording* page of the Sipelia Plugin role. Changing this setting at the entity level forces the entity to no longer inherit the value from the global setting, thus allowing you to turn recording on or off for only specific entities, without affecting all of them.

9 Click **Apply**.

10 Add more SIP intercoms by repeating the steps above.

Example

Watch this video to learn more. Click the **Captions** icon (**CC**) to turn on video captions in one of the available languages.

After you finish

- Register your SIP intercom.
- Associate a Security Center entity with your SIP intercom.

Related Topics

2N Integration and Installation Manual

Associating Security Center entities with SIP intercoms

To confirm caller identification through live video and grant access to the premises through access-controlled doors, you can associate various Security Center entities to a SIP intercom, and then control everything through Security Desk.

Before you begin

Add and configure your SIP intercom in Config Tool.

What you should know

- With SIP intercoms, you can associate cameras, doors, zones, and device outputs along with their respective output behaviors.
- When you associate Security Center entities, the icons that appear in the Sipelia™ conversation window indicate the action that can be performed on the entity. The icons do NOT indicate the current state of the entity. For example, if a door is associated with the intercom, an "unlock" icon ( ) that appears in the conversation window indicates that the door is currently locked, and you can unlock the door by clicking the icon.

Procedure

1 From the Config Tool homepage, open the *Area view* task.

2 From the Area view, select a SIP intercom, and then click the **VoIP** tab.

3 In the **Associated entities** section, click **Add an entity** ( ).

4 In the *Entity association wizard*, select one of the following entities, and then click **Next**. Follow the onscreen instructions to complete the entity association.

- **Camera:** The camera that you want to associate with the SIP entity. You can associate a camera so that the live video stream from this Security Center camera is displayed during active phone calls between SIP entities.
- **Door:** The door that you want to associate with the SIP intercom. You can associate a door so that you can unlock it to grant access to callers that use the intercom, and then lock it once the callers have entered.
- **Zone:** The zone that you want to associate with the SIP intercom. You can associate a zone so that you can arm and disarm it based on the callers that use the intercom. As a result, you can grant or deny callers access to a particular section of your premises. Once set up, in the Security Desk conversation window or tile, click to disarm the zone.
- **Device:** The output of a device that you want to associate with the SIP intercom. Each output must have an associated output behavior; for example, sounding a buzzer (through an output relay) when a window that is equipped with a glass break sensor (connected to an input) is shattered. Once set up, you can trigger the selected output behavior during a call.

5 If you selected a device output, do the following to define the output behavior:

a) In the **Alias** field, enter a short, descriptive name that allows you to quickly identify the output behavior.

b) Select an applicable output type (*Normal*, *Active*, or a custom one). You can create multiple custom output types based on the *State*, *Pulse*, and *Periodic* types. For more information on output behaviors, see About output behaviors.

6 Once the *Entity association wizard* has closed, click **Apply** to save your changes.

The entity that you selected appears in the list of associated entities.

7 (Optional) If your SIP intercom has a built-in camera, and you want the video stream from this camera to be displayed in conversations involving the intercom, you can associate that built-in camera by clicking .

8 (Optional) To configure the associated entity that you added, click .

9 (Optional) Associate custom events to call states or Sipelia entity states.

10 Repeat the above steps to associate more entities to the SIP intercom.

11 For multiple entities of the same type, use the arrow buttons in the *Associations* panel to define the order in which the entities appear within the conversation window or tile in Security Desk. For example, if you have associated multiple cameras, the live video stream from the camera that appears first in the list is the default stream. Users can switch to the video streams of other cameras by clicking , and then selecting another camera.

If you associate all of the possible entities with a SIP intercom, all of the entities appear within the conversation window or tile in Security Desk. As shown in the following image, each entity can be controlled during an active call.



- **A:** Trigger the output behavior of a device
- **B:** Arm and disarm a zone
- **C:** Open and close a door
- **D:** View the video stream of the selected camera

Example

Let's assume that you want to facilitate how operators respond to lost card requests. With a SIP intercom installed at the main entrance of the building, you can associate the following two entities with that intercom: the camera at the main entrance, and the door at the main entrance. When Charles calls from that SIP intercom, claiming that he has lost his card and cannot enter the building, you can confirm Charles' identity by comparing the live video against his cardholder profile, and then granting or denying access to the building accordingly.

Watch this video to learn more. Click the **Captions** icon (**CC**) to turn on video captions in one of the available languages.

Recording the audio and video of call sessions

To record the audio and video of call sessions involving users and SIP intercoms, you must configure the recording settings provided in the *Recording* page of the Sipelia™ plugin role.

Before you begin

Make sure that you have advanced licenses installed on your system.

What you should know

The **User recording** and **Device recording** options apply to all user and SIP intercom entities unless an entity is configured differently in which case it will not inherit the default values from the Sipelia plugin role anymore.

Procedure

1 From the Config Tool homepage, open the *Plugins* task.

2 Select the Sipelia plugin role, and then click **Recording**.

3 Enter the following:

- **User recording:** Enables the recording of call sessions in which user entities participate (either as a caller or recipient of a call). Once recorded, call sessions can later be reviewed and exported through the Call report task in Security Desk. The default value is defined here at the role level, and is inherited by all user entities. It is possible to turn the recording on or off for only specific user entities via the **Recording audio and video** setting provided in the *VoIP* page of the user entity without affecting all of them.
- **Device recording:** Enables the recording of call sessions in which the SIP intercom entities participate (either as a caller or recipient of a call). Once recorded, sessions can later be reviewed and exported through the Call report task. The default value is defined here at the role level, and is inherited by all SIP intercom entities. It is possible to turn the recording on or off for only specific SIP intercom entities via the **Recording audio and video** setting provided in the *VoIP* page of the SIP intercom entity without affecting all of them.
- **Recording folder:** The directory in which the recorded files are stored. It can be a local or a network directory. The Sipelia Plugin role displays an error when the path format is not valid, the local path is not accessible, or the network path does not exist or is unreachable. If a directory specified with a local path does not exist, it will automatically be created. If a path becomes invalid, the call session recording will stop and the recorded files will be lost.
- **Automatic cleanup:** Enables automatic deletion of the recorded files. This option is enabled by default with a retention period of 30 days.

4 Click **Apply**.

After you finish

If deploying Sipelia, configure SIP accounts for Security Desk users.

Related Topics

Recording public address calls on page 59

Registering your SIP intercom with Sipelia Server

To make calls from your SIP intercom to Security Center users, you must register the intercom with Sipelia™ Server.

Before you begin

- Install the SIP intercom according to the recommendations of the intercom manufacturer.
- Add and configure your SIP intercom in Config Tool.

What you should know

There are many SIP intercoms that you can install, and the way they are configured and registered to Sipelia Server might differ. The steps below provide a general overview of the settings that must be configured. Always refer to the documentation provided by the manufacturer of the SIP intercom for information on how to configure and register the intercom.

To register your SIP intercom with Sipelia Server:

- Create a SIP account on your intercom.
- Enter an applicable name for your SIP account.

  This name can be the same as the one given to the intercom when adding it to Security Center, but it does not need to be. The SIP account name is not used during SIP communications.
- Enter the domain or IP address of Sipelia Server.

  You can find the IP address of Sipelia Server in **Config Tool** > **Network view** > **Properties**.
- For the SIP port value, enter the value that you configured. The default value used by the *Session Initiation Protocol (SIP)* is **5060**.
- Enter the SIP extension that was assigned to this intercom in Security Center. In some SIP clients, you must enter the extension number as the username.
- Enter the password for the SIP extension that has been assigned to the intercom.
- Register the intercom so that it can communicate with Sipelia Server.

To authenticate incoming calls from your SIP intercom:

- Calls can be authenticated upon request by enabling authentication in the **Server** tab of the Sipelia plugin role.

  When **Enable authentication on incoming calls** is enabled, authentication is required for all incoming calls and user credentials are verified.

The SIP intercom is ready to make and receive calls.

After you finish

To create ring groups that include other ring groups and SIP intercoms, create ring groups.

Viewing the inventory of VoIP entities

Using the VoIP inventory reporting task, you can view a list of all SIP entities and all Security Center users with SIP accounts in your system. You can use the information in this report to manage your system and to troubleshoot issues.

What you should know

- To update the report after it has been generated, click **Refresh** ( ).
- You can edit an entity directly from the report by selecting the entity and clicking **Go to the configuration page of this entity** ( ).
- A **Save** button ( ) is included in the VoIP inventory task, however, the XML file that is exported is only used by Genetec Inc. technical support personnel.

Procedure

1 From the Config Tool homepage, open the *VoIP inventory* task.

2 From the **Filter by type** drop-down, select which entities you want to include in the report.

The entities and their properties are listed in the report pane. You can use the information in the report to manage your system and to troubleshoot issues. For example, you can see in the report if any SIP users do not have an assigned extension.

Report pane columns for the VoIP inventory task

After generating a report, the results of your query are listed in the report pane. This section lists the columns available for the VoIP inventory task.

- **Type:** Type of entity.
- **Name:** Name of the entity.
- **Extension:** Extension assigned to the entity.
- **Server:** Name of the server to which entity is connected. *Local* or a *trunk* server.
- **IP address:** IP address of the intercom or the Security Desk Sipelia™ client.
- **Connections:** The number of SIP connections.
  - **Intercom:** Indicates how many SIP clients are registered as extensions of the intercom entity.
  - **User:** Indicates how many Security Desk clients the user is currently logged into.
- **Attached entities:** Number of attached entities for the intercom or user.
- **Recording:** A check mark indicates that communication from this entity is recorded.
- **First registration:** The first time that the user or intercom registered to the Sipelia Server.
- **Latest registration:** The most recent time that the user or intercom sent a *keepalive* message to the Sipelia Server.
- **Up time:** The time difference between the intercom or user's *first registration* and *latest registration*.
- **State:** The current state of the intercom (*available* or *offline*) or the user (*available*, *away*, *busy*, or *offline*).
- **Priority:** The call priority that is assigned to the user.
- **Gateway enabled:** Indicates if the user profile has Sipelia Gateway enabled.

Associating Security Center custom events with Sipelia entity states and call states

You can associate custom events to Sipelia™ entities, for example to monitor when a user registers to your SIP server or a device goes offline. You can also associate custom events to call states to be notified, for example, when a call was missed or to monitor errors on intercom devices.

Before you begin

- Create the Security Center custom events that you want to associate with entity states or call states. For more information, see Creating custom events on the TechDoc Hub.
- To monitor the state of SIP intercoms, configure the registration timeout delay for your intercoms.

What you should know

Any type of custom event from Security Center can be associated to a Sipelia entity state or to a call state. A call state represents the different phases of a call, for example that the call is ringing, busy or in progress. The entity state shows when the entity is online or offline. Two types of entities can be associated to custom events: users and intercoms.

You can associate custom events in the following way:

- **Call states:** Call states can be associated with users, local intercoms (intercoms connected directly to the Sipelia Server, not to a trunk), and with intercoms connected to a trunk.
- **Entity states:** Entity states can be associated with users and local intercoms only. To get the state of an intercom connected to a trunk, you need to monitor the state of the trunk itself.

TIP: You can search and combine events from different Sipelia entities in a single report using the *Unified report* investigation task. Using this data, you can create dashboards to monitor the information that matters to you. For more information, see Investigating record types and About dashboards on the TechDoc Hub.

Procedure

To associate a Security Center custom event with the state of a Sipelia entity or the state of a call:

1 Open Config Tool and choose one of the following:
   - **For users:** Open the **User Management** task and select the user you want to associate with the event.
   - **For intercoms:** Open the **Area view** task and select the intercom you want to associate with the event.

2 Click the **VoIP** tab to access the Sipelia parameters.

To associate a custom event to a call state:

1 In the *Custom events association* section, under the *Call state events* list, click **Add an item** ( ).

2 In the *Assign custom event to call state* window, select the call state you want from the **Call state** drop-down:
   - **Ringing:** A call to or from an intercom or user is ringing.
   - **On a call:** A call with an intercom or user is established.
   - **Busy:** A call with an intercom or user cannot be established for any of the following reasons:

- The intercom or user is already in a call with another participant.
- The intercom or user rejects the call.
- The intercom is connected through a trunk server and the trunk server is not able to reach this intercom.

- **Not answered:** The intercom or user does not accept or reject the call. The call stops after the ringing timeout.
- **Error:** Unexpected errors occurred during the call.
- **End call:** A call was established without error and is properly closed by any of the participants.

3 Select the required custom event from the list and click **Add**.

4 Click **Apply** to confirm your selection.

The associated event now appears in the Call state events list.

To associate a custom event to an entity state:

1 In the *Custom events association* section, under the *Entity state events* list, click **Add an item** ().

2 In the *Assign custom event to entity state* window, select the entity state you want to associate from the **Entity state** list:

- **Online:** The entity first registers or registers back after being offline.
- **Offline:** The entity fails to register or to re-register within the configured period of time.

3 Select the required custom event from the list and click **Add**.

4 Click **Apply** to confirm your selection.

The associated event now appears in the Entity state events list.

The custom events that you have selected are now linked to the Sipelia entity or call states of your choice. You can associate an action to these events, monitor them, or include these events in a report.

Configuring Sipelia role failover

You can protect the Sipelia™ role against hardware failures by assigning standby servers to the role.

Before you begin

- Your SIP devices (intercoms, SIP phones, etc.) need to be able to connect to two SIP servers to get failover on Sipelia. Verify with your manufacturer that your devices support that configuration type.
- You require a centralized SQL database.
- Install Sipelia Server on a main and at least one expansion server to use as a standby server for the Sipelia role.
- The Sipelia servers need access to a shared database and a shared network drive.

Procedure

1 From the Config Tool homepage, open the *Plugins* task.

2 Select the Sipelia role ( ) and click **Resources**.

3 Under the Servers list, click **Add an item** ( ) and select a server where *Sipelia Server* has been installed.

4 If you want the primary server to take control of the role after it is restored, select the **Force execution on highest priority server** option.

By default, the role remains on the secondary server after a failover occurs to minimize system disruptions.

5 Configure the Sipelia database to be shared with both the primary and secondary servers.

The database is set up when you create the plugin role.

6 Configure the recording folder of each server to point to the shared network drive.

After you finish

You can register your SIP devices to the primary and the secondary SIP servers that you configured. See your manufacturer's user guide to get detailed information about your SIP device configuration.

Ring groups

A ring group is a group of SIP entities that has its own unique SIP phone extension. All entities (or members) within a ring group are part of a call list, and all members get called when the ring group extension is called.

In Security Center, a ring group can include any combination of the following entities: users, user groups, nested ring groups, and SIP devices. The members of a ring group can either be called all at once, or successively at a set interval. The call stops ringing when any one of the members within a call list answers the call.

NOTE: Security Center user groups in a ring group do not need unique extensions, but each user entity included in the ring group must provide one. If users and SIP devices do not have a SIP extension, they are bypassed when the ring group that they are a part of is called.

User ring groups

A user ring group is a Security Center user group that can only include Security Center users and other Security Center user groups.

A user ring group has the following characteristics:

- It can be assigned its own unique SIP extension.
- To receive a call, users that are part of a user ring group must have their own dedicated SIP extension.
- When the extension is dialed, all members of the user group set with a SIP extension are called.

Creating ring groups

To call multiple SIP entities at the same time, you can create a *ring group* in Config Tool.

Before you begin

Depending on which entities you want to include in your ring group, do one or more of the following:

- Configure SIP accounts for your Security Center users.
- Create a user ring group.
- Add your SIP intercoms.

What you should know

A ring group can include any combination of the following entities: users, user groups, nested ring groups, workstations, and SIP devices.

NOTE: You can configure more than one level for a nested ring group, but all entries under the nested ring group are called as an **All at once** ring group.

Procedure

1 From the Config Tool homepage, open the *Plugins* task.

2 Select the Sipelia™ plugin role, and then click **Ring groups**.

3 Click **Add ring group**.

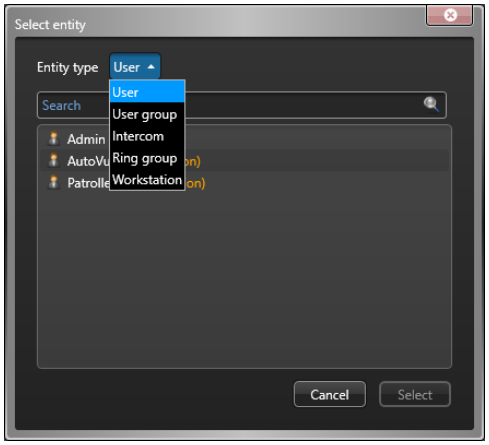
4 Enter a descriptive name for your ring group, and then click **Add**.

5 Assign a SIP extension to your ring group in one of the following ways:

- Click **Auto-assign**. Choose an existing range, and then click **Apply**. This is the recommended approach.
- Enter the **SIP extension**. Enter the SIP entity's phone extension manually.

6 In the **Call list** field, click **Add entity**.

7 In the *Select entity* dialog box, choose the different entities that you want to include in your ring group. You can use the **Entity type** drop-down list to filter by entity type, and then search for the entities by name.



8 Select all of your entities, and click **Select**.

9 (Optional) To configure the associated entity that you added, click .

10 In the **Call recipients** field, select one of the following:

- **All at once:** All members of a call list are called at the same time. The call stops ringing when any one of the members within a call list answers the call.
- **Successively every:** Members are called in sequence, with a set delay between each call, in the order that they appear in the call list. This sequence of calls is repeated until any one of the members answers the call. If a member declines the call, the next member is called immediately rather than after the set delay. The minimum delay is 10 seconds. Because it might affect how long a call to a ring group goes unanswered, it is not recommended to set a high delay.

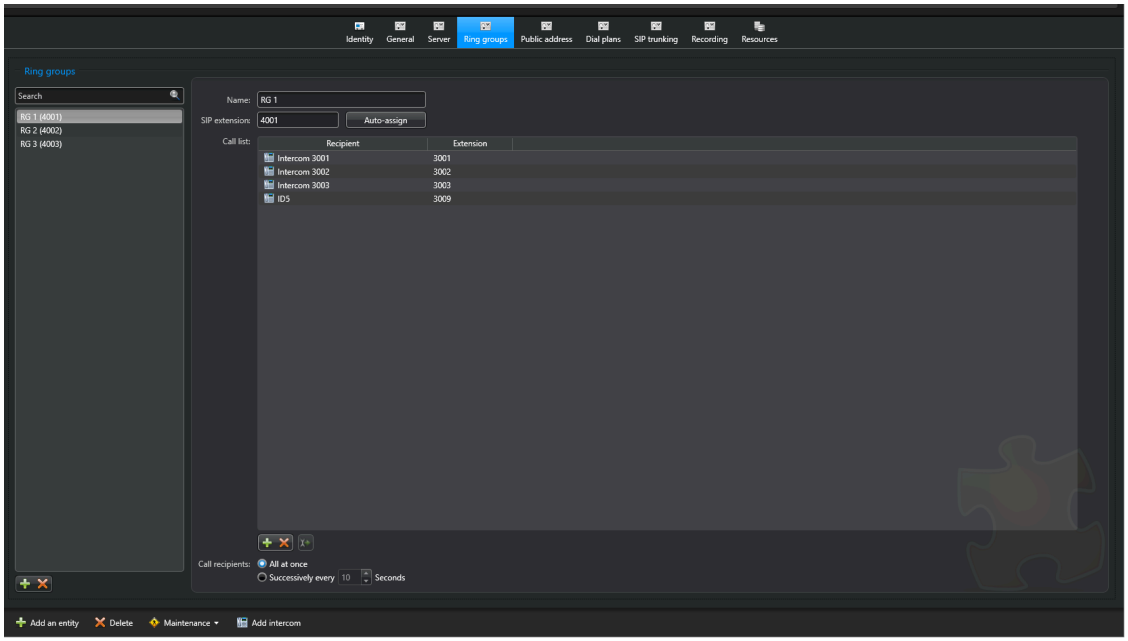
11 (Optional) To change the order in which the members of the ring group are called, use the arrow buttons to move the members up or down.

This is available only when **Successively every** is selected.

12 Click **Apply**.

Example

The image below shows a ring group with three entities: one user, one user group, and one user ring group. The call sequence for this ring group is set at *Successively every 10 seconds*. When the ring group extension (5001) is called, Charles' extension (6001) is called first. If Charles does not answer or decline the call within 10 seconds, each of the users in the *Operators* user group who already have an assigned extension is called. The same sequence continues until one of the members in the call list answers the call.



Watch this video to learn more. Click the **Captions** icon (**CC**) to turn on video captions in one of the available languages.

After you finish

If deploying Sipelia, install Sipelia Client.

Creating user ring groups

To call multiple Security Center users at the same time, you can create a user ring group by assigning a SIP extension to a Security Center user group.

Before you begin

Configure SIP accounts for your Security Center users.

What you should know

A user ring group is a Security Center user group that can only include Security Center users and other Security Center user groups.

Procedure

1 From the Config Tool, the open*User Management* task.

2 Click **User groups**, and then select one from the list.

3 Create a Security Center user group that contains all the users that you want in your ring group. For information on creating user groups in Config Tool, see the *Security Center Administrator Guide*.

4 Click the **VoIP** tab.

The members of the user group appear in the **Call list** field.

5 Assign a SIP extension to your ring group in one of the following ways:

- Click **Auto-assign**. Choose an existing range, and then click **Apply**. This is the recommended approach.
- Enter the **SIP extension**. Enter the SIP entity's phone extension manually.

6 In the **Call recipients** field, select one of the following:

- **All at once:** All members of a call list are called at the same time. The call stops ringing when any one of the members within a call list answers the call.
- **Successively every:** Members are called in sequence, with a set delay between each call, in the order that they appear in the call list. This sequence of calls is repeated until any one of the members answers the call. If a member declines the call, the next member is called immediately rather than after the set delay. The minimum delay is 10 seconds. Because it might affect how long a call to a ring group goes unanswered, it is not recommended to set a high delay.

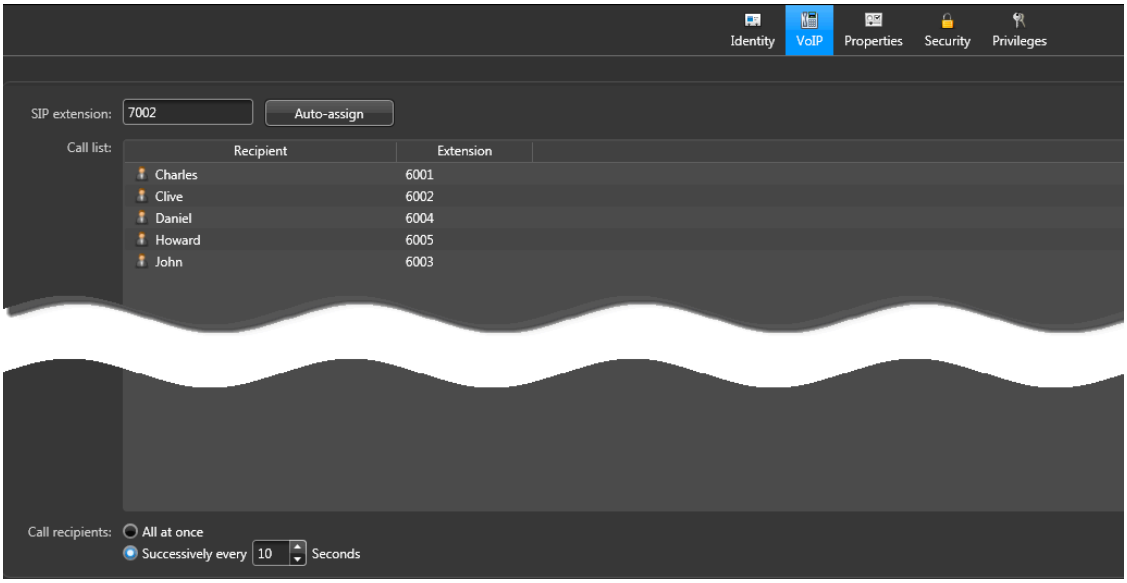
7 (Optional) To change the order in which the members of the ring group are called, use the arrow buttons to move the members up or down.

This is available only when **Successively every** is selected.

8 Click **Apply**.

Example

The image below shows a user ring group with five Security Center users, each with their own dedicated extensions. The call sequence for this ring group is set at *Successively every 10 seconds*. When the ring group extension (7002) is called, Charles' extension (6001) is called first. If Charles does not respond within 10 seconds, Clive's extension (6002) is called. The sequence continues until one of the members answers the call.



Watch this video to learn more. Click the **Captions** icon (**CC**) to turn on video captions in one of the available languages.

After you finish

If deploying Sipelia™, add your SIP intercoms.

Configuring Sipelia Client

To participate in voice and video calls, you can connect the required audio and video equipment to the Security Desk workstations onto which Sipelia™ Client is installed, and then configure relevant configuration settings for each Security Center user.

Before you begin

- Configure SIP accounts for your Security Center users.
- Install Sipelia Client on each of the Security Desk workstations that run Sipelia.
- Install the required headsets and webcams. For optimal audio quality, it is recommended to use headsets instead of microphones and speakers.

Procedure

1 Sign in to Security Center with Security Desk.

2 Click **Options** > **Sipelia**.

3 In the *Audio and video* section, set the following audio and video devices that are used for calls.

- **Webcam:** The webcam you want to use for video calls. The webcam must be connected to the Security Desk workstations that run Sipelia.
- **Microphone:** The microphone you want to use for calls. The microphone must be connected to the Security Desk workstations that run Sipelia.
- **Speaker:** The speaker you want to use for calls. The speaker must be connected to the Security Desk workstations that run Sipelia.
- **Secondary ringer:** You can set a ringer to play in a separate output from the main speaker.

IMPORTANT: Make sure that the devices are properly configured in the Windows operating system.

NOTE:

- In cases where both sides of the communication use a different codec, communication is still possible as Sipelia Server transcodes the media stream.
- From the Quality drop-down, you can select the streamed webcam video quality. By default, the quality is set to *high*, however, you can lower the video quality if your system has network bandwidth restrictions or if there are restrictions from other communication endpoints in the system.

4 Click to expand the *Advanced* section, and set the following settings, as required:

- **Video codecs:** The video codecs that are supported by Security Desk for video communication. By default, the H.264 and H.263 codecs are turned on, and should suffice for most cases. As a result, it is recommended to keep the default settings, and to be aware that changing video codecs can disrupt the video that is streamed during video calls.
- **UDP port range:** The port range for the User Diagram Protocol (UDP). The UDP ports are used by the different SIP clients to send and receive communication data. The default range is from **20000** to **20500**. It is recommended to keep the default settings, and to change them only if Sipelia logs any port-related issues about making or receiving calls with Security Desk.

5 Set the following call-related options, as required:

- **Open new calls in:** Select whether you want all incoming calls to automatically open in the conversation window or in a tile within the *Monitoring* task in Security Desk.
- **Do not disturb:** Select this option to reject incoming calls automatically when on a call.
- **Automatic forward:** Select this option to automatically forward a call to a different extension.

NOTE: You must have call forwarding permissions and be logged in to forward a call.

- **Show keypad on call:** Select this option to automatically display the keypad on call start-up.
- **Ringtone:** Use the **Test** button to test the current *Default* ringtone. Use the **Volume** slider to set the ringtone volume for incoming calls.

  NOTE: The **Volume** slider changes the volume of the ringtone, it doesn't affect the volume of the voice call.

6 Repeat these steps on each of the Security Desk workstations that run Sipelia.

Example

Watch this video to learn more. Click the **Captions** icon (**CC**) to turn on video captions in one of the available languages.

After you finish

- Test the audio and video devices by making voice and video calls between SIP clients.
- If deploying Sipelia, configure two-way communication between Sipelia Server and other SIP servers.

Related Topics

Changing the ringtone for incoming Sipelia calls on page 46

Changing the ringtone for incoming Sipelia calls

You can select a new WAV audio file to be used as the ringtone for Sipelia™ calls. The ringtone can be customized for each Security Desk workstation.

Before you begin

Ensure that the new ringtone audio file you want to use is in WAV format and is accessible from the client machine.

What you should know

In **Security Desk > Options > Sipelia**, the **Ringtone** drop-down list displays the *Default* ringtone. You cannot add new ringtones to the drop-down list, but you can save a new WAV file as the *Default* ringtone.

Procedure

1 On the machine where Sipelia Client is installed, open File Explorer, and navigate to *Computer\Local Disk (C:)\Program Files (x86)\Genetec Sipelia\Sounds*.

2 Create a backup copy of the existing *ringtone.wav* and save under a new filename.

3 Save the new ringtone WAV file you want to use as *ringtone.wav*.

4 Log on to Security Desk.

NOTE: If you are already logged on, you do not need to log off/log on nor do you need to restart the application.

5 Click **Options** > **Sipelia**.

6 Click the **Test** button which is located next to the **Ringtone** drop-down list.

The system plays the new ringtone you saved as *ringtone.wav*.

7 Move the **Volume** slider to set the ringtone volume.

NOTE: The **Volume** slider changes the volume of the ringtone, and does not affect the volume of the voice call.

Configuring two-way communication between Sipelia Server and other SIP servers

To expand your SIP capabilities, you can configure two-way communication between Sipelia™ Server and other SIP servers, so SIP extensions from either server can call each other.

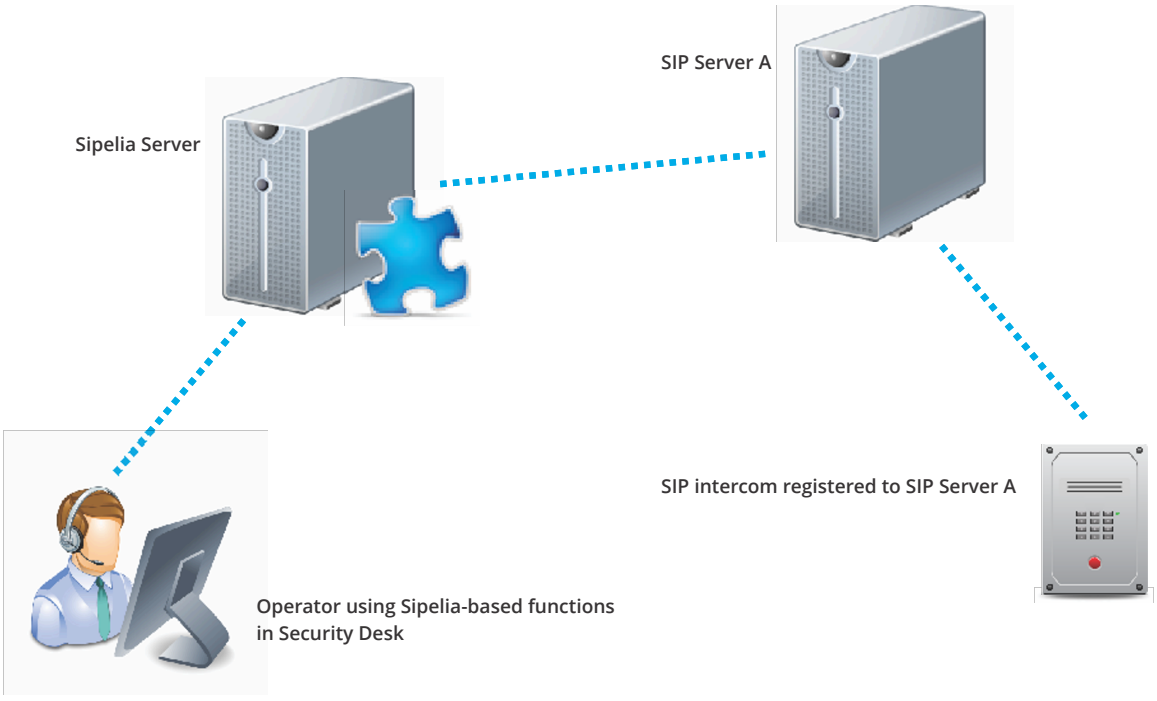
Before you begin

- Read the documentation provided by the manufacturer of the SIP server that you want to connect to Sipelia Server.

  NOTE: For the purpose of this topic, we will refer to the SIP server that you want to connect to as *SIP Server A*.
- Make sure that *SIP Server A* supports trunking and is able to connect to other SIP servers (in our case, Sipelia Server). Some SIP servers do not support trunking, and therefore, cannot call extensions registered to other SIP servers.
- Make sure that *SIP Server A* has been certified by Genetec Inc. as a compatible hardware component, and that it is included in Sipelia's supported hardware list for SIP servers and intercom servers.

What you should know

To have a SIP environment like the one shown in the following image, you must properly configure two-way communication between the two SIP servers (Sipelia Server and *SIP Server A* which, for example, is embedded within an intercom server). In this illustration, Sipelia Server must add *SIP Server A* as a SIP trunk, and *SIP Server A* also needs to add and configure Sipelia Server as a SIP trunk. Also, each SIP server must define appropriate dial plans to be able to contact the other SIP server.



Because there are many different SIP servers available on the market, and numerous possible implementations for each, the method used to configure a SIP trunk and dial plan process differs for each SIP server. As a result, the steps listed below only provide general instructions on how to set up a two-way communication between Sipelia Server and *SIP Server A*. However, in this topic, and in the related topics about SIP trunks and dial plans, specific instructions are provided for how to set up a one-way communication

between Sipelia Server and the SIP trunk for *SIP Server A*. This way, extensions on Sipelia Server can call those on *SIP server A*.

Procedure

To connect Sipelia Server to *SIP Server A*:

1 Add *SIP Server A* as a SIP trunk.

2 Define dial plan rules that allow SIP extensions registered to Sipelia Server to call SIP extensions registered to *SIP Server A*.

3 Import the dial plan rules in Config Tool.

To connect *SIP Server A* to Sipelia Server:

1 Add Sipelia Server as a SIP trunk. Refer to the documentation of *SIP Server A* for details on how to add SIP trunks.

2 Define and implement dial plan rules that allow SIP extensions registered to *SIP Server A* to call SIP extensions registered to Sipelia Server. Refer to the documentation of *SIP Server A* for details on how to define and implement dial plan rules.

Configuring your SIP intercom to call a specific extension

When your SIP intercom can only be configured to call a specific extension when a button is pressed, it is recommended to configure your SIP intercom to call the SIP extension of a ring group, so that you can manage the list of users who will receive the call.

Before you begin

- Install the SIP intercom according to the recommendations of the intercom manufacturer.
- Add and configure your SIP intercom in Config Tool.

What you should know

When your SIP intercom only allows the configuration of one SIP extension, it is recommended that you use the extension of a ring group. With a ring group you can then decide who will be called when the button is pressed on the SIP intercom and it calls the extension. The ring group also provides you with the flexibility to adjust the list of recipients to meet your needs.

Because there are a variety of SIP intercoms that you can install, the way they are configured and registered to Sipelia™ Server might differ. The steps below provide a general overview of the settings that must be configured. Always refer to the documentation provided by the manufacturer of the SIP intercom for information on how to configure and register the intercom.

Procedure

1 Add and configure a ring group in Config Tool.

2 Take note of the SIP extension that you assigned to the ring group.

3 Create a SIP account on your intercom.

4 Enter an applicable name for your SIP account. This name can be the same as the one given to the intercom when adding it to Security Center, but it does not need to be. The SIP account name is not used during SIP communications.

5 Enter the domain or IP address of Sipelia Server.

You can find the IP address of Sipelia Server in **Config Tool** > **Network view** > **Properties**.

6 For the SIP port value, enter the value you have configured. The default value used by the *Session Initiation Protocol (SIP)* is **5060**.

7 Enter the SIP extension which was assigned to the ring group in Security Center. In certain SIP clients, you must enter the extension number as the username.

The SIP intercom will call the extension of the ring group when its button is pressed. This means that one or more recipients could be called all at the same time or one after another according to the ring group configuration.

Adding SIP intercom objects to a map

You can add SIP intercom objects to your maps so that they are displayed in Security Desk and can be used by the operators to make and receive calls.

What you should know

Maps are enabled in Security Center 5.4 and later.

Procedure

1 From the Config Tool homepage, open the *Map designer* task.

2 Select a map.

3 Navigate to the location on the map where you want to add the SIP intercom object.

4 From the **Entities** list in the toolbar, select **Area view** ( ).

5 From the **Area view** list, drag the SIP intercom to the required location on the map.

6 Adjust the size, position, and orientation of the SIP intercom object with the mouse. From the panel displayed on the right, you can make precise adjustments to the size and position coordinates of the SIP intercom icon.

7 Click **Apply**.

The SIP intercom now appears on the map in the Security Desk *Maps* task. Make a test call to the SIP intercom to test functionality.

Enabling and Disabling codecs on Sipelia servers

To change the default codecs that are allowed to run on Sipelia™ servers, you must enable or disable them through a configuration file.

Before you begin

- Create a backup of the following file:

  C:\ProgramData\Genetec Sipelia\SipServer\SipServer.config

What you should know

- Sipelia supports several audio and video codecs.
- Enabling and disabling video and audio codecs is done through an XML configuration file. In this file, you can find the following XML elements: **AudioCodecs** and **VideoCodecs**. The **Codec** element within them contains the **Enabled** attribute that defines the state of each codec.

Procedure

To enable or disable a specific audio or video codec:

1 Deactivate the Sipelia plugin role (see *Security Center Administrator Guide*).

2 Open the following configuration file:

C:\ProgramData\Genetec Sipelia\SipServer\SipServer.config

3 In the chosen **Codec** element, modify the **Enabled** attribute to `False` to disable the codec or to `True` to enable it.

CAUTION: You must only change the **Enabled** attribute, never the **Name** attribute. It is not possible to add new types of codecs directly by editing this configuration file.

4 Save the file and reactivate the Sipelia plugin role.

To restore default values without a backup file:

1 Deactivate the Sipelia plugin role.

2 Delete the configuration file:

C:\ProgramData\Genetec Sipelia\SipServer\SipServer.config

3 Reactivate the Sipelia plugin role.

The configuration file will be regenerated with the default values.

Related Topics

High-quality audio codecs on page 52

High-quality audio codecs

Audio quality in Sipelia™ depends on the audio codec used by the communication parties.

For new Sipelia installations, high-quality audio is available by default for communication between Security Desk workstations.

To enable high-quality audio for an intercom, the intercom must support at least one of the high-quality codecs supported by Sipelia.

When upgrading a Sipelia system, to enable high-quality audio, you must manually change the Sipelia Server codec configuration. This is necessary because the Sipelia upgrade process does not make any changes to the existing configuration.

Supported high-quality audio codecs:

- L.16-44-1
- Speex (Ultrawideband)
- Speex (Wideband)
- Opus

Related Topics

Supported codecs on page 12
Enabling and Disabling codecs on Sipelia servers on page 51

Configuring the device registration margin for SIP intercoms

SIP intercoms send regular *keepalive* messages to the Sipelia™ Server to indicate that they are still running and connected. The Sipelia Server knows how often to expect a *keepalive* message based on the *registration timeout* which is sent from each SIP intercom. You can configure a *device registration margin* as a safeguard for SIP intercoms that might exceed their *registration timeout* period.

Before you begin

- Create a backup of the following file:

  C:\ProgramData\Genetec Sipelia\SipServer\SipServer.config
- Investigate the *registration timeout* for the SIP intercoms in the system. Depending on the devices you have connected, this setting might be configurable or hardcoded.

What you should know

- SIP intercoms can sometimes send their *keepalive* message shortly after their *registration timeout* period has elapsed. When this happens, Security Center reports that the device is fluctuating between online and offline states. This issue can be solved by increasing the *device registration margin* which adds additional time to the device's *registration timeout* period.
- The default value of the *device registration margin* is 60 seconds. The minimum accepted value is 0 seconds.

Procedure

1 Deactivate the Sipelia plugin role (see *Security Center Administrator Guide*).

2 Open the following configuration file:
C:\ProgramData\Genetec Sipelia\SipServer\SipServer.config

3 To adjust the *device registration margin*, locate the `DeviceRegistrationMargin` element and modify the value (in seconds).

4 Save the file and reactivate the Sipelia plugin role.

Configuring the SIP trunk state timeout period

The Sipelia™ Server monitors the SIP Trunks to ensure that they are still running and connected. You can modify how often the Sipelia Server monitors the SIP Trunk through a configuration file on the Sipelia Server.

Before you begin

- Create a backup of the following file:

  C:\ProgramData\Genetec Sipelia\SipServer\SipServer.config

What you should know

- Unlike SIP intercoms, SIP Trunks do not send a keepalive message to the Sipelia Server. The Sipelia Server checks the status of SIP Trunks using the configurable *trunk state timeout period*.
- The default value of the *trunk state timeout* is 60 seconds. The minimum accepted value is 40 seconds.

Procedure

1 Deactivate the Sipelia plugin role (see *Security Center Administrator Guide*).

2 Open the following configuration file:
C:\ProgramData\Genetec Sipelia\SipServer\SipServer.config

3 Locate the `TrunkStateTimeout` element and modify the value (in seconds).

4 Save the file and reactivate the Sipelia plugin role.

Configuring DTMF sequences

You can configure DTMF sequences in Sipelia™ and send or receive the configured DTMF sequences in the call's *Conversation* window.

What you should know

- After you configure DTMF sequences, they will be accessible from a *Call* window. You can then click a button on the keypad to send a series of DTMF tones such as *123*.

Procedure

1 From the Config Tool homepage, open the *Plugins* task.

2 Select the Sipelia plugin role, and then click **Server**.

3 In the *DTMF sequences* section, click **Add** ( ) to add a new sequence.

4 Enter a name, a tone and a description.
   a) **Name**: Not mandatory, maximum 250 characters.
   b) **Tones**: Limited size. Can only contain the following characters : 0-9, #, *.
   c) **Description**: Not mandatory, maximum 250 characters.

5 When a sequence is selected, you can.
   a) Click Delete to delete this tone from the list.
   b) Click **Edit** ( ) to edit this tone.
   c) Click the green/red light to activate or deactivate this sequence. A deactivated sequence will not be available for the call.

   The sequence is now added to the list.

Sending DTMF sequences

Once you configure the DTMF sequences, you can display the list of available and active DTMFs and send them through a call.

Before you begin

Configure DTMF sequences

Procedure

1 Launch a call to another entity. Typically, an intercom or a different user.

2 Press **Keypad** ( ) to display the list of available and active DTMF sequences.

3 Click on a DTMF sequence to start sending it through the call.

4 When you are sending a DTMF sequence, all other sequences and dialpad numbers are *Disabled* until the sequence has finished sending.

Receiving DTMF sequences

After you configure the DTMF sequences, active sequences can be received by Security Desk users during a call.

Before you begin

Configure DTMF sequences

Procedure

1 Launch a call between a Security Desk user and another entity. Typically, an intercom or a different user.

2 The other entity sends a DTMF sequence.

3 An event is triggered when the received sequence matches a configured sequence.

Adding a workstation

You can add a workstation using Config Tool. After a workstation is added in Sipelia™ it is available in your list of contacts.

What you should know

- You can call a workstation regardless of which user is logged on, because you direct the call to a workstation and not a specific user.
- If the same user is logged into multiple workstations, we recommend enabling the **Limit concurrent logons** in the **User management** task. For more information, refer to the *Security Center Administrator Guide*.

Procedure

To add a workstation:

1 From the Config Tool homepage, open the **Plugins** task.

2 Select the Sipelia plugin role, and then click **Server**.

3 In the *Workstations* section, click **Add** ( ).

4 Enter a name and an extension:

- **Name**: Not mandatory, maximum 250 characters.
- **Extension**: Not mandatory, maximum 250 characters.

To assign a workstation:

1 Click **Add**, and then click **Apply**.

The workstation is added to your contact list.

2 To assign and link the workstation to Sipelia, select the workstation from the drop-down list in the *Call* dialog box in Security Desk.

The workstation is **Assigned** as a callable workstation.

To unassign a workstation:

1 From the Sipelia plugin role, click **Server**.

2 In the *Workstations* section, click **Unassign workstation** to remove the link to Sipelia.

3 In the *Workstations* section, you can also do one of the following:

- Click **Delete** ( ) to delete this workstation from the list.
- Click **Edit** ( ) to edit this workstation.

NOTE: You need Administrator privileges to unassign a workstation.

Creating paging zones

To call multiple SIP entities at the same time with a public address, you can create a paging zone in Config Tool.

What you should know

- A paging zone can include any combination of users, intercoms, and paging zones.
- Entities without extensions cannot be added to a paging zone. An entity is removed from a paging zone if their extension is deleted.

Procedure

1 From the Config Tool homepage, open the *Plugins* task.

2 Select the Sipelia™ plugin role, and then click **Public Address**.

3 Click **Add a paging zone**.

4 Enter a descriptive name for your paging zone, and then click **Add**.

5 Enter a description for the paging zone.

6 Assign an extension to your paging zone:

- Click **Auto-assign**. Select a range, and click **Apply** to assign the next available extension in the selected range.

  This is the recommended approach.
- Enter the **Extension** manually.

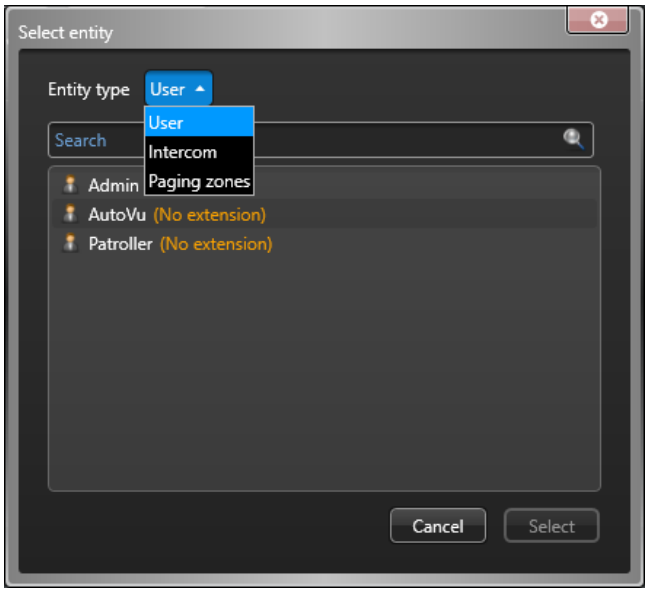
7 Enter the **Automatic start timeout**, to define the amount of time, in seconds, until the broadcast begins.

Devices can connect before the message starts. If all the devices in the paging zone accept the call, the message starts even if the timeout is not complete. If a device accepts after the timeout, it can join the paging group with the message in progress.

8 Click **Add a paging zone member**.

9 In the *Select entity* dialog box, choose the entities that you want to include in your paging zone.

You can use the **Entity type** list to filter by entity type, and then search for the entities by name.



10 Select an entity and click **Select**.

11 To add an audio message, click **Add audio message** and browse to the file.

Ensure that the audio files are in the correct format. The supported file formats are *WAV* and *MP3*.

12 Click **Apply**.

The **Audio files folder** displays the path where the audio files are saved. When using failover, the location should point to a shared network folder that is accessible by all expansion servers.

IMPORTANT: Sipelia does not offer the ability to adjust the timing of messages between multiple speakers. If this adjustment is required, refer to our Supported Device List for products that offer this feature.

Recording public address calls

You can enable audio recording for a specific paging zone to record public address calls. The recorded audio is available in the *Call report* task.

Before you begin

Create paging zones

Procedure

1 From the Config Tool homepage, open the *Plugins* task.

2 Select the Sipelia™ plugin role, and then click **Public Address**.

3 Select a zone and enable **Record audio**.

You can only record live messages.

Related Topics

Recording the audio and video of call sessions on page 33

Broadcasting messages

You can broadcast dynamic messages using the public address feature. You can choose to broadcast live, text-to-speech, or pre-recorded messages.

Before you begin

- Create paging zones with extensions.
- Log into Security Desk with a user that has a VoIP SIP extension assigned.

Procedure

To broadcast a message live:

1 From the Security Desk homepage, open the address book and launch a paging zone.

2 In the **Public address setup** window, select **Microphone** as the broadcast source.

3 Use the headset connected to the Security Desk workstation and click **Broadcast** to talk live.

Your voice broadcasts on all the speakers that are connected to the paging zone.

To broadcast a message from text:

1 From the Security Desk homepage, open the address book and launch a paging zone.

2 Open the address book and launch a paging zone.

3 In the **Public address setup** window, select **Text to speech** as the broadcast source.

4 Write the text message and click the plus sign to **Add a message**.

It cannot contain special characters.

NOTE: You can switch languages using the Windows language feature and link multiple languages for the same message.

5 You can select the **Repeat** icon to loop the message up to three times.

6 Click **Broadcast** to start the public address call.

To broadcast a pre-recorded message:

1 From the Security Desk homepage, open the address book and launch a paging zone.

2 Open the address book and launch a paging zone.

3 In the **Public address setup** window, select a pre-recorded audio file.

Files that were previously uploaded on your Sipelia server are available as *MP3* or *WAV* files. The selected file plays on the speakers connected to the paging zone.

4 Click **Broadcast** to play the audio file.

Automating public announcements

You can trigger pre-recorded or text-to-speech actions using event-to-actions.

Procedure

To set events for public announcements.

1 From Config Tool, open the System task and click the **General settings** view.

2 Click **Actions**.

3 Click **Add an item** ( ).

4 From the **When** list in the *Event-to-action* dialog box, select the event.

5 From the **From** list, select the entity.

6 Select the **Action** and attributes for each type of event.

- For a pre-recorded message, select **Trigger Public Announcement [pre-recorded]** and then select the audio message.
- For a text message, select **Trigger Public Announcement [text-to-speech]** and then enter the text message and language.

7 Select the paging zone and click **Save**.

To set an event to load an attached camera in a tile.

1 From Config Tool, open the System task and click the **General settings** view.

2 Click **Actions**.

3 Click **Add an item** ( ).

4 From the **When** list in the *Event-to-action* dialog box, select the event.

5 From the **From** list, select the entity.

6 From the **Action** list, select **Display caller attached camera in a tile** to load the attached camera in a Security Desk tile.

7 Select the recipient and click **Save**.

Configuring STUN Server in Sipelia

You can configure Sipelia™ to use a STUN server using the *SipServer.config* file.

Procedure

1 Deactivate the Sipelia Plugin role.
In the *Plugins* task, right-click the Sipelia Plugin and select **Maintenance** > **Deactivate role**.

2 Open the Sipelia Server configuration file (*C:\ProgramData\Genetec Sipelia\SipServer\SipServer.config*).

3 Change the value of the `NatTraversalMethod` property to *STUN*.
Example:

```
NatTraversalMethod Value="STUN" StunServerAddress="stun.l.google.com:19302"
 AutoDetect="True"/
```

4 Configure the server and the port.

5 Activate the Sipelia plugin role.
In the *Plugins* task, right-click the Sipelia Plugin and select **Maintenance** > **Activate role**.
NOTE: Security Desk Sipelia clients automatically use the same STUN configuration as the server.

Configuring SRTP in Sipelia

You can configure Sipelia™ to enable SRTP using the *SipServer.config* file.

What you should know

The SRTP setting should be used with TLS encryption enabled to secure the connection and the data exchange between the server and the client. The available modes are:

- None
- Prefer (recommended protocol if supported)
- Force (forces the server to use SRTP)

Procedure

1. Deactivate the Sipelia Plugin role.
   In the *Plugins* task, right-click the Sipelia Plugin and select **Maintenance** > **Deactivate role**.
2. Open the Sipelia Server configuration file (*C:\ProgramData\Genetec Sipelia\SipServer\SipServer.config*).
3. Change the value of the `SrtpMode` to *Force*.
4. Activate the Sipelia plugin role.
   In the *Plugins* task, right-click the Sipelia Plugin and select **Maintenance** > **Activate role**.
   NOTE: Security Desk Sipelia clients automatically use the same SRTP setting as the server.

Configuring the TURN server in Sipelia

To enable communication between the Sipelia™ Gateway role and mobile devices or web browsers that are behind network address translators (NATs) or firewalls, you can configure the Traversal Using Relays around NAT (TURN) server in Sipelia.

Before you begin

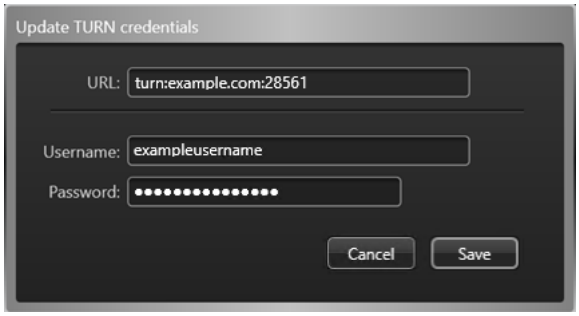
You must have a TURN server subscription from a third-party service provider.

Procedure

1 From the Config Tool homepage, open the *Plugins* task.

2 Select the Sipelia Gateway role and click the **Properties** tab.

3 Click **Update TURN credentials**.
The *Update TURN credentials* window opens.



4 Enter the following credentials:

- **URL:** Address of the TURN server for relaying data when direct connections aren't possible due to NAT or firewall restrictions. The URL must start with `TURN:`.
- **Username:** Enter the value provided by the third-party TURN server provider.
- **Password:** Enter the value provided by the third-party TURN server provider.

5 Click **Save**.

4

SIP trunks and dial plans

This section includes the following topics:

- "Adding SIP trunks" on page 65
- "Associating Security Center custom events with SIP trunk states" on page 67
- "Dial plans" on page 68
- "Creating dial plans in Config tool" on page 72
- "Dial plan scenario 1: Forwarding to a SIP trunk all calls starting with a prefix" on page 74
- "Dial plan scenario 2: Reserving a range of SIP extensions for local calls" on page 76
- "Dial plan scenario 3: Reserving a range of SIP extensions for calls to a SIP trunk" on page 79
- "Dial plan scenario 4: Replacing source SIP extensions" on page 82
- "Dial plan scenario 5: Removing prefix on source SIP extensions from a SIP trunk" on page 84
- "Dial plan scenario 6: Forwarding calls to another SIP extension on schedule" on page 86
- "About importing dial plans" on page 89

Adding SIP trunks

To connect to SIP servers other than Sipelia™ Server, you can add a SIP trunk in Config Tool, and then define applicable dial plan rules to make and receive calls between the SIP servers.

Before you begin

Configure the time interval that Sipelia uses to check SIP trunk status.

What you should know

- SIP trunks work together with dial plans. For example, to connect Sipelia Server to another SIP server (*SIP Server A*) through a SIP trunk, you must define dial plan rules that instruct Sipelia Server about which calls to reroute through the trunk and on towards the new SIP extension destinations that are registered to *SIP Server A*. Furthermore, if you want proper two-way communication between Sipelia Server and SIP Server A, both SIP servers need to be configured accordingly.
- When Sipelia Server is connected to another SIP server (*SIP Server A*) through a SIP trunk, *SIP Server A* can request authentication from the Sipelia Server. The Sipelia Server must confirm its identity using the outbound credentials configured for *SIP Server A*.
- Each trunk can have its own outbound credentials configured in **Outbound requests**.
- When inbound authentication is enabled between trunks, you must ensure that the caller extension is configured as an intercom belonging to another Server. Otherwise, the Sipelia Server will revert to its default authentication setting.

Procedure

1 From the Config Tool homepage, open the *Plugins* task.

2 Select the **Sipelia**™ role, and then click the **SIP trunking** tab.

3 Click **Add SIP trunk**.

4 In the **Name** field, enter a descriptive name for your SIP trunk.

The name of the SIP trunk must be unique because it will be used in the dial plan rules involving this SIP trunk.

Example: You might name the SIP trunk *TrunkSIPServerA* for *SIP Server A*.

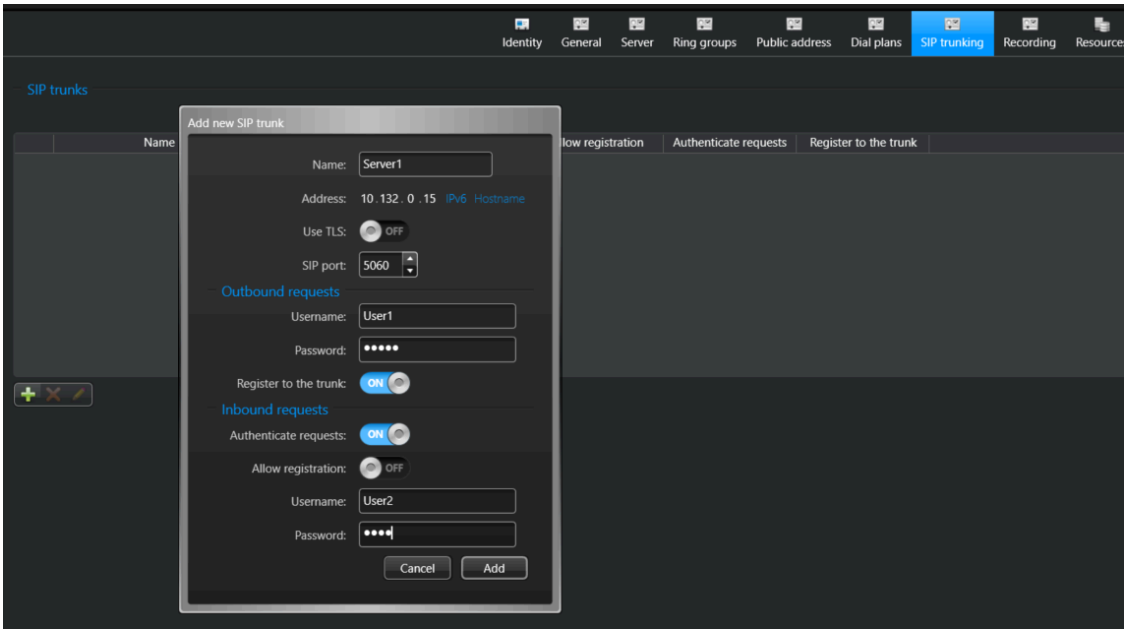
5 Complete the following fields:

- **IP address:** The IP address of the SIP server that you want to connect Sipelia™ Server to.
- **SIP port:** The port used by the SIP trunk to communicate with the Sipelia™ Server. Because SIP trunks are SIP servers, the default value is **5060**.

  NOTE: A Sipelia Server IP address might be: **10.150.4.100** with port **5060**.
- **Use TLS:** When selected, the SIP connection to the trunk server goes through TLS transport.
- **Outbound requests:** The credentials used by the Sipelia Server when authenticating to another Server.
- **Register to the trunk:** When selected, the Sipelia Server will try to register the trunk.
- **Inbound requests:** The credentials used by the Sipelia Server to validate the identity of another server.
- **Authenticate requests:** When selected, incoming requests from the trunk are challenged for authentication.

6 Click **Add**, and then click **Apply**.

The SIP trunk *TrunkSIPServerA* has been added. The LED in the first column indicates the state of the trunk: green for online and red for offline.



IMPORTANT: The SIP extension configured for an intercom entity in Security Center is the unique identifier of this entity. For intercoms connected to an Intercom Exchange Server connected with a trunk to Sipelia, this SIP extension must be used to call the intercom. This extension must also be the identifier used by the intercom to call Sipelia clients. The Dial Plan rules must be properly configured for these extensions to match.

After you finish

To call extensions registered on *TrunkSIPServerA*, define the applicable dial plan rules.

Associating Security Center custom events with SIP trunk states

You can associate custom events to SIP trunk states to monitor when the connection with the trunk is established, for example, or goes offline.

Before you begin

- Create the Security Center custom events that you want to associate with a trunk state. See the *Security Center Administrator Guide, Creating custom events* for details.
- Configure the time interval Sipelia™ uses to check SIP trunk status.
- Add your SIP trunks.

What you should know

Any type of custom events from Security Center can be associated to a SIP trunk state. The trunk states show when it is online or offline.

Procedure

1 Open Config Tool and select the **Plugins** task.

2 Select the **SIP trunking** tab of the Sipelia plugin.

3 Click on the trunk that you want to monitor.

The *Trunk state event* section appears on the page.

4 Click **Add an item**( ) under the *Trunk state events* control.

5 In the *Assign custom event to trunk state* window, select the state you want from the **Trunk state** drop-down list:

- **Online:** The connection to the trunk is available when the periodic check is performed.
- **Offline:** There is no connection with the trunk when the periodic check is performed.

6 Select the required custom event from the *Custom event* list and click **Add**.

7 Click **Apply** to confirm your selection.

The associated event now appears in the Trunk state events list.

The custom event you have selected is now linked to the SIP trunk state of your choice. You can now associate an action to this event, monitor it or include this event in a report. You can also add more events, delete them or modify them by using the buttons under the events list.

Dial plans

A dial plan is a collection of rules that defines how calls are routed locally or between two SIP trunks. Dial plans ensure that calls are routed and rerouted correctly, and they also allow administrators to block calls to certain geographic locations or ensure the privacy of the callers.

A dial plan defines one or multiple rules to specify the following:

- How calls can reach SIP extensions that reside on other SIP servers.
- How calls coming from other SIP servers can reach SIP extensions that reside on Sipelia™ Server.
- How calls can be forwarded even when they remain local to Sipelia Server.
- How Sipelia Server can modify dialing information such as source or destination SIP extensions during call routing.

For example, if a SIP extension registered to Sipelia Server needs to call an extension that resides on *SIP Server A*, you must first configure a SIP trunk in Sipelia Server to connect to *SIP Server A*. Then you must define a dial plan to route the call from Sipelia Server to *SIP Server A*. In Sipelia, dial plans can also be used to route local calls, that is, calls that must remain in Sipelia Server.

NOTE: Dial plans also need to be configured on *SIP Server A* to route calls from and to Sipelia Server. Refer to the documentation provided by the manufacturer of *SIP Server A* for information on how to configure dial plans.

Defining dial plan rules

To call SIP extensions registered to SIP servers other than Sipelia™ Server, or to redirect calls from within Sipelia Server, you must define dial plan rules.

Before you begin

- If you want to call SIP extensions that are registered to other SIP servers, add a SIP trunk for that SIP server.
- Familiarize yourself with *dial plans* and dial plan rules used in Sipelia.

What you should know

Rules defined in a dial plan are listed in order of priority. A rule located in a previous row is considered to be a higher priority rule. If a call matches two or more rules, the first rule listed in the dial plan will always apply. The order of priority can be changed after they were imported in Config Tool using the arrows located on the right side of the list.

Procedure

1 Create a new text file for your dial plan with the *.txt* or *.csv* extension, or open an existing one.

2 Write an applicable dial plan rule.

3 Enter a value for each field required by the rule.

IMPORTANT: A dial plan rule must contain the correct number of comma-separated values for it to be imported successfully. Each comma-separated value of a rule must be separated by a semicolon (**;**).

4 Repeat the same steps for additional rules that you require in your plan.

5 Save and close the dial plan file.

After you finish

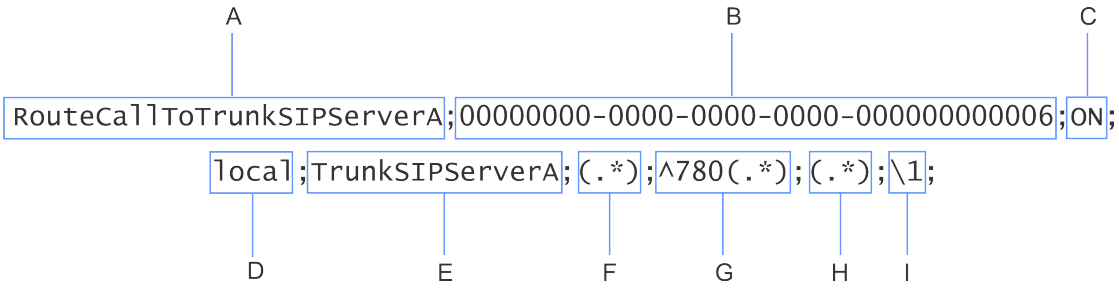
Import your dial plan rule in Config Tool.

Dial plan rules

Dial plan rules can be added, edited, and deleted in Config Tool or imported from comma-separated values (CSV) text files, where they are stored. The text files can contain multiple rules. Each rule is on a separate line and each comma-separated value (or value field) of a rule is separated by a semicolon (**;**). Sample dial plan rules are added by default as inactive.

Although CSV files support different types of separators (or delimiters), dial plan rules in Sipelia™ only support the semicolon (;) separator. You can define dial plan rules for a number of reasons. Most commonly, dial plan rules are used to call SIP extensions that are registered to SIP servers other than Sipelia Server, and to redirect local calls from extensions residing within Sipelia Server.

Each dial plan rule must define the values as shown in the following example.



- **Name (A):** The name of the dial plan rule. If the name is not provided, a default value is used upon import (*DialplanRule1*). Enter a name that will help you remember the purpose of the rule.
- **Schedule (B):** The schedule that defines when this rule must be in effect. The long string of digits for this value represent a GUID (Globally Unique Identifier). If this GUID is not provided, or it is incorrect, the default value of *Always* is used upon import.

  TIP: The schedule can be changed graphically from the *Dial plans* page in Config Tool once the dial plan rule has been imported, so you can leave it empty here and let the system assign the default schedule.
- **Status (C):** The status of the rule (On = Active; Off = Inactive). This value is not case sensitive. If this value is not provided, the default value *Off* is used upon import.
- **Direction from (D):** The SIP server from where the call is originating (the source server). Possible values are *local* or the name of a SIP trunk. Sipelia Server is always the *local* SIP server. If a SIP trunk name is entered here and does not exist in the Sipelia Server configuration, a warning will be displayed, and this value will revert to *local* during import.
- **Direction to (E):** The SIP server that receives the call (the destination server). Possible values are *local* or the name of a SIP trunk. The name of the SIP trunk must be unique, and must match the name that was entered in the *SIP trunking* tab when adding the SIP trunk. Sipelia Server is always the *local* SIP server. If a SIP trunk name is entered here and does not exist in the Sipelia Server configuration, a warning will be displayed, and this value will revert to *local* during import.

  TIP: After adding the SIP trunk in Config Tool, you can copy its name by right-clicking on the name. It is recommended to copy the name, and then paste it within your dial plan rule to avoid misspelling it.
- **Source (F):** The regular expression that identifies the extension of the caller (the source caller). The dial plan rule is only applied if there is a match between the regular expression and the caller's extension. This field is mandatory.
- **Destination (G):** The regular expression that identifies the extension of the recipient (the destination recipient). The dial plan rule is only applied if there is a match between the regular expression and the recipient's extension. This field is mandatory.
- **New source (H):** The value that changes the caller's extension (if the rule is applied). For example, if the caller at extension 1001 calls the recipient at extension 2001, and the *New source* value is 4001, the recipient at extension 2001 sees that the incoming call is coming from extension 4001, not 1001. Possible values can also include regular expressions.
- **New destination (I):** The SIP extension that is receiving the call (if the rule is applied). Possible values can also include regular expressions.

Order of priority for dial plan rules

Rules defined in a dial plan are listed in order of priority. A rule located in a previous row is considered to be a higher priority rule. If a call matches two or more rules, the first rule listed in the dial plan will always apply. The order of priority can be changed after they were imported in Config Tool using the arrows located on the right side of the list.

Sample scenarios

For more information on how to use regular expressions in dial plan rules, you can refer to the following sample scenarios for some of the more common uses for dial plans:

- Forwarding to a SIP trunk all calls starting with a prefix.
- Reserving a range of SIP extensions for local calls.
- Reserving a range of SIP extensions for calls to a SIP trunk.
- Replacing source SIP extensions.
- Removing the prefix on source SIP extensions from a SIP trunk.
- Forwarding calls to another SIP extension on schedule.

In the Sipelia installation folder (typically *C:\Program Files (x86)\Genetec Sipelia*), the *Samples* folder contains a ZIP file from which you can extract the sample dial plans for common scenarios.

Regular expressions in Sipelia

A regular expression is a sequence of symbols used by a regular expression engine to identify all the strings of characters that match a specific search pattern without having to list all the possible discrete values that must be returned. The Microsoft's .NET Framework Regular Expression engine is the engine used in Sipelia™.

In Sipelia™, regular expressions are used in dial plan rules to do the following:

- Look for specific SIP extensions from which calls are made.
- Look for destination SIP extensions to which calls are made.
- Look for specific prefixes added to SIP extensions.
- Change SIP extensions from which calls are received.
- Change SIP extensions to which calls are sent.

Regular expression elements

Regular expressions used in dial plan rules typically include the following elements:

- **(.*):** Match any value.
  - Use this element in the **Source (F)** or **Destination (G)** fields to request Sipelia Server to look for any source or destination extension when calls are made.
  - Use this element in the **New source (H)** or **New destination (I)** fields to ensure that the source or destination extension will remain unchanged when routed by Sipelia Server.
- **\n:** Match the value of a capturing group from a previous related field. A capturing group is one or more regular expression elements that are typically specified between parentheses and match a specific pattern.
  - Use this element in the **New source (H)** field to specify the capturing group from the *Source* field to be used as the new source value. **n** represents the ranking of the capturing group (\1; \2; \5, etc.) of the

regular expression. For example, if *Source* contains **550[1-5](.*)**, enter \2 to use **(.*)** as the *New source* value, which in this case will remove the prefix **550** and the following digit.

- Use this element in the **New destination (I)** field to specify the capturing group from the *Destination* field to be used as the new destination value.

- **n:** Match a specific value. In Sipelia, this value generally represents a specific SIP extension.
  - Use this element in the **New source (H)** field to specify a different extension number from which the calls are made. This is useful if you want calls to come from an extension which differs from the original source extension.
  - Use this element in the **New destination (I)** field to specify the extension number that will receive the calls. This is useful if you want to forward calls to an extension which differs from the original destination extension.
- **[first - last]:** Match any single character in the range from **first** to **last**.
- **{n}:** Match the previous element exactly **n** times.
- **^:** Match value at the beginning of the string.
- **\b:** Use at the beginning and at the end of a series of regular expression elements to match on whole word only (not only a part of it).

Example

For example, the regular expression **\b6[0-9]{2}\b** could be used to look for SIP extensions 600 to 699.

Regular expression construct	Description
\b	SIP extension boundary. Used together with the same **\b** element at the end of the expression, this requires the whole SIP extension to be matched. Starting or ending characters will not be omitted.
6	Look for a SIP extension that begins with **6**.
[0-9]	Look for the digits **0** through **9**.
{2}	Look for 2 occurrences of the above digits following the **6**.
\b	Match a SIP extension boundary.

For more information on regular expressions, see Microsoft's website.

Creating dial plans in Config tool

Sample dial plan rules are added to Sipelia™ by default as inactive. You can add or edit the rules directly in the Dial plan Config Tool page.

Procedure

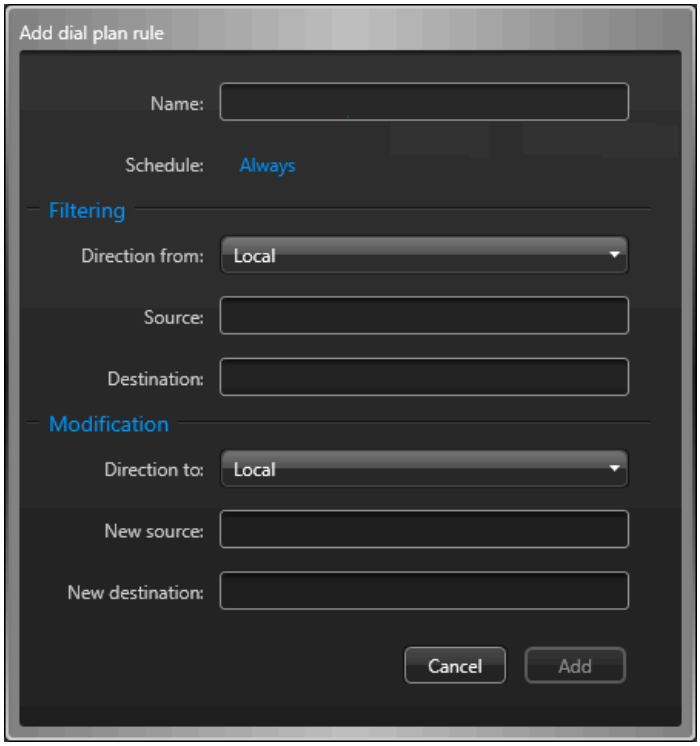
To add a dial plan:

1 From the Config Tool homepage, open the *Plugins* task.

2 Select the **Sipelia**™ plugin role, and then click **Dial plans**.

3 Click ( ).

4 In the **Add dial plan rule** window, enter a value for each field required by the rule.



- **Filtering**: Defines the rule attributes that filter the calls. For example, you can select to which call the rule should be applied by setting the **Direction from**, **Source** and **Destination** fields.
- **Modification**: Defines the rule attributes that modify the call. For example, you can alter the call that was selected using the **Filtering** attributes. You can set the following fields **Direction to**, **New source** and **New destination** fields.

5 Click **Add**.

To edit an existing dial plan:

1 From the Config Tool homepage, open the *Plugins* task.

2 Select the **Sipelia**™ plugin role, and then click **Dial plans**.

3 Select a dial plan rule in Config Tool.

4 Click ( ).

5 In the **Edit dial plan rule** window, enter a new value for each field required by the rule.

6 Click **OK**.

After you finish

For more information on dial plan rules refer to the following sample scenarios:

- Forwarding to a SIP trunk all calls starting with a prefix.
- Reserving a range of SIP extensions for local calls.
- Reserving a range of SIP extensions for calls to a SIP trunk.
- Replacing source SIP extensions.
- Removing the prefix on source SIP extensions from a SIP trunk.
- Forwarding calls to another SIP extension on schedule.

Dial plan scenario 1: Forwarding to a SIP trunk all calls starting with a prefix

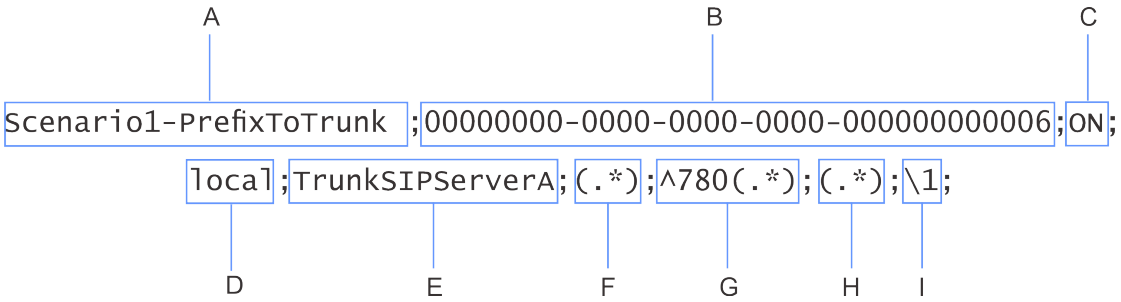
In this scenario, a dial plan rule is used to route to a SIP trunk any call that begins with a specific prefix.

Scenario

- **What do you want to do?** Your SIP extension is registered to *Sipelia™ Server* and you need to connect to *SIP Server A* because you want to call SIP extensions (SIP client, SIP intercom, etc.) that are registered to that SIP trunk. Doing so allows you to establish a one-way communication with *SIP Server A*.
- **What you need to do before defining the rule?** Add a SIP trunk in Config Tool for *SIP Server A* and name it *TrunkSIPServerA*. This trunk name must be unique, so that the dial plan rule that you will create will not conflict with other rules, and your rule will be applied correctly.
- **How can you do it?** To route calls to *TrunkSIPServerA*, you can use a dialing prefix; for example, the prefix **780**. This means that SIP extensions on Sipelia™ Server must dial this prefix if they want to call SIP extensions on *TrunkSIPServerA*. The prefix approach mimics that of dialing 9 to get an outside line in a typical PBX system. The prefix can be any number that you want, as long as it is defined in your dial plan rule.

Example of the dial plan rule

The following is an example of a dial plan rule that accomplishes the above mentioned scenario.



The value labels identified below correspond to the column labels which appear on the *Dial plans* page of the **Sipelia** plugin role.

Value letter	Value label	Description
A	**Name**	The name of the rule indicates that calls made from Sipelia™ Server and starting with a prefix will be routed to the SIP trunk.
B	**Schedule**	The schedule is set to *Always*, meaning that the rule will always be verified.
C	**Status**	The status is set *On*, meaning that the rule is currently active.
D	**Direction from**	The field is set to *local* to look for calls that will be originating from Sipelia Server.
E	**Direction to**	The field is set to *TrunkSIPServerA* because calls will be routed to the SIP trunk.

Value letter	Value label	Description
F	**Source**	The regular expression is set to **(.*)**, meaning that the call can be made from *any* SIP extension registered to Sipelia Server. If you choose to enter a specific SIP extension, make sure that the regular expression corresponds to the caller's extension.
G	**Destination**	The prefix **^780** is used. This means that SIP extensions on Sipelia Server must first dial 780 to be able to communicate with *TrunkSIPServerA*. Furthermore, the prefix is followed by the regular expression **(.*)** This regular expression creates a regular expression capturing group for all digits that follow the prefix. **Example**: If you want to call extension 1001 on *TrunkSIPServerA*, you must dial **7801001**. The regular expression **(.*)** creates a group index of extensions that begin with *1*. As a result, the extension 1001 is included in this index. If you call **7801002**, the extension 1002 will also be included in this index.
H	**New source**	The regular expression **(.*)** is used. This means that the caller's extension on Sipelia Server remains unchanged. **Example**: If the source extension registered on Sipelia Server is 6001, the dial plan rule will not cause it to change. With this rule, the same applies to all other registered extensions on Sipelia Server. However, if you enter a *New source* value of 7001, and you make a call from your extension (6001), the recipient sees that the incoming call is coming from extension 7001, not 6001.
I	**New destination**	Because the *New destination* value is linked to the *Destination* field, the **\1** indicates that the rule must use the value of the first regular expression group of *Destination*. In this scenario, the *Destination* value is **780(.*)**, so using **\1** will correspond to **(.*)** which is actually the destination SIP extension without the prefix.

Result

Once this dial plan rule is imported into Config Tool, if any SIP extension registered to Sipelia Server dials **7801001**, the extension **1001** on *SIP Server A* will ring.

Dial plan scenario 2: Reserving a range of SIP extensions for local calls

In this scenario, dial plan rules are used to define a range of SIP extensions for calls that remain local to Sipelia™ Server, while calls made to any other SIP extensions are automatically routed to a SIP trunk.

Scenario

- **What do you want to do?** You want the SIP extensions 4000 to 4500 reserved for calls that must remain local to Sipelia Server, while calls made to other SIP extensions are routed to *SIP Server A*.
- **What you need to do before defining the rule?** Add a SIP trunk in Config Tool for *SIP Server A* and name it *TrunkSIPServerA*. This trunk name must be unique, so that the dial plan rule that you will create will not conflict with other rules, and your rule will be applied correctly.
- **How can you do it?** You need to define two separate rules listed in the right order in your dial plan.
  - The first rule will keep the calls local when the destination SIP extensions are between 4000 and 4500.
  - The second rule will take calls to any other SIP extensions and route them to *TrunkSIPServerA*.

IMPORTANT: The rules need to be in the right order in the dial plan to work as described in this scenario.

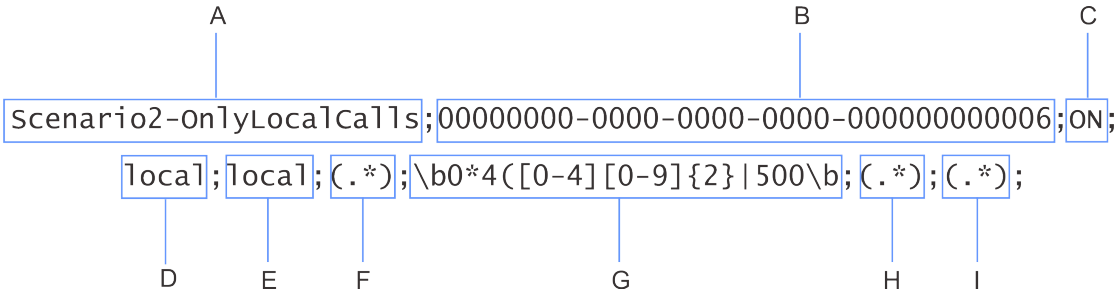
Example of the dial plan

The following is an example of a dial plan that accomplishes the above mentioned scenario.

1. Scenario2-OnlyLocalCalls;00000000-0000-0000-0000-000000000006;ON;local;local;(.*);\b0*4([0-4][0-9]{2}|500)\b;(.*);(.*);
2. Scenario2-RestCallsToMyTrunk;00000000-0000-0000-0000-000000000006;ON;local;TrunkSIPServerA;(.*);(.*);(.*);(.*);

Rule 1: Only local calls

The first rule listed in the dial plan and shown below tells Sipelia Server to look for source SIP extensions between 4000 and 4500, and keep them local.



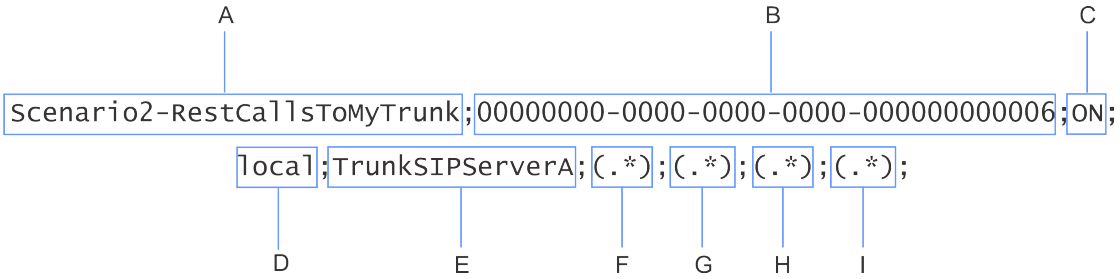
The value labels identified below correspond to the column labels which appear on the *Dial plans* page of the **Sipelia** plugin role.

Value letter	Value label	Description
A	**Name**	The name indicates that this rule will route local calls.
B	**Schedule**	The schedule is set to *Always*, meaning that the rule will always be verified.

Value letter	Value label	Description
C	**Status**	The status is set *On*, meaning that the rule is currently active.
D	**Direction from**	The field is set to *local* to look for calls that will be originating from Sipelia Server.
E	**Direction to**	The field is set to *local* because the calls that match the rule will remain local to Sipelia Server.
F	**Source**	The regular expression is set to **(.*)**, meaning that the call can be made from *any* SIP extension registered to Sipelia Server.
G	**Destination**	The regular expression is set to **\b0*4([0-4][0-9]{2}|500)\b** to look for destination SIP extensions between 4000 and 4500, as described below: • **\b:** SIP extension boundary. Used together with the same **\b** element at the end of the expression, this requires the whole SIP extension to be matched. Starting or ending characters will not be omitted. • **0*:** Look for any number of zeros before the next character, which is **4**. • **4:** Look for a SIP extension that begins with **4**. • **([0-4]:** Look for a single occurrence of digits **0** through **4** after the first **4**. • **[0-9]{2}:** Look for 2 following occurrences of digits **0** through **9**, which now covers extension 4000 to 4499. • **|500):** Look specifically for **500** to add 4500 in the search pattern. • **\b:** Match a SIP extension boundary.
H	**New source**	The regular expression **(.*)** is used. This means that your extension (the source caller) on Sipelia Server remains unchanged.
I	**New destination**	The regular expression **(.*)** is used. This means that the called extension remains unchanged.

Rule 2: Other calls routed to the SIP trunk

The second rule listed in the dial plan and shown below tells Sipelia Server to route to *TrunkSIPServerA* any call that is not matching the first rule.



Value letter	Value label	Description
A	**Name**	The name indicates that this rule will route any other call to *TrunkSIPServerA*.
B	**Schedule**	The schedule is set to *Always*, meaning that the rule will always be verified.
C	**Status**	The status is set *On*, meaning that the rule is currently active.
D	**Direction from**	The field is set to *local* to look for calls that will be originating from Sipelia Server.
E	**Direction to**	The field is set to *TrunkSIPServerA* to route the calls to that SIP trunk.
F	**Source**	The regular expression is set to **(.*)**, meaning that the call can be made from *any* SIP extension registered to Sipelia Server.
G	**Destination**	The regular expression is set to **(.*)**, meaning that the call can be made to reach *any* SIP extension.
H	**New source**	The regular expression **(.*)** is used. This means that your extension (the source caller) on Sipelia Server remains unchanged.
I	**New destination**	The regular expression **(.*)** is used. This means that the called extension remains unchanged.

Result

Once the dial plan is imported into Config Tool, only the highest priority rule will apply. This dial plan results in the following:

1. If any SIP extension dials a number between 4000 and 4500, the call will remain local to Sipelia Server.
2. If any SIP extension dials any other number, the call will automatically be routed to *SIP Server A*.

Dial plan scenario 3: Reserving a range of SIP extensions for calls to a SIP trunk

In this scenario, dial plan rules are used to define a range of SIP extensions for calls that are automatically routed to a SIP trunk.

Scenario

- **What do you want to do?** You want to reserve SIP extensions 4000 to 4500 for calls that must automatically be routed to *SIP Server A*.
- **What you need to do before defining the rule?** Add a SIP trunk in Config Tool for *SIP Server A* and name it *TrunkSIPServerA*. This trunk name must be unique, so that the dial plan rule that you will create will not conflict with other rules, and your rule will be applied correctly.
- **How can you do it?** You need to define two separate rules listed in the right order in your dial plan.
  - The first rule will take any call made to a SIP extension that is between 4000 and 4500 and automatically route it to *TrunkSIPServerA*.
  - The second rule will take calls to any other SIP extension and will keep it local to Sipelia™ Server.

IMPORTANT: The rules need to be in the right order in the dial plan to work as described in this scenario.

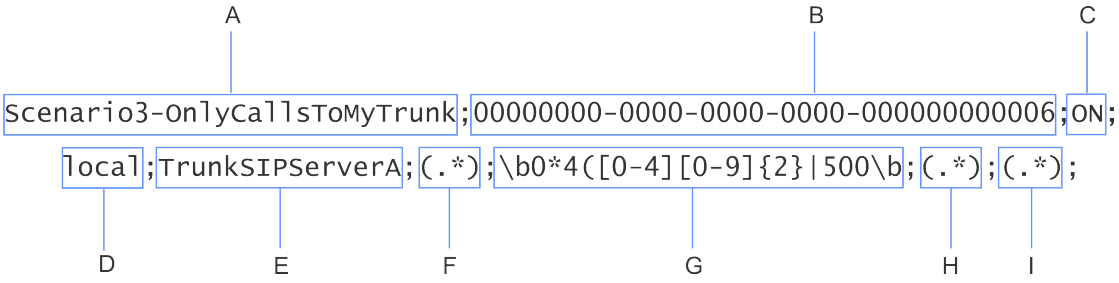
Example of the dial plan

The following is an example of a dial plan that accomplishes the above mentioned scenario.

1. Scenario3-OnlyCallsToMyTrunk;00000000-0000-0000-0000-000000000006;ON;local;TrunkSIPServerA;(.*);\b0*4([0-4][0-9]{2}|500)\b;(.*);(.*);
2. Scenario3-RestAreLocalCalls;00000000-0000-0000-0000-000000000006;ON;local;local;(.*);(.*);(.*);(.*);

Rule 1: Calls to the SIP trunk only

The first rule listed in the dial plan and shown below tells *Sipelia™ Server* to look for destination SIP extensions between 4000 and 4500, and forward them to the SIP trunk.



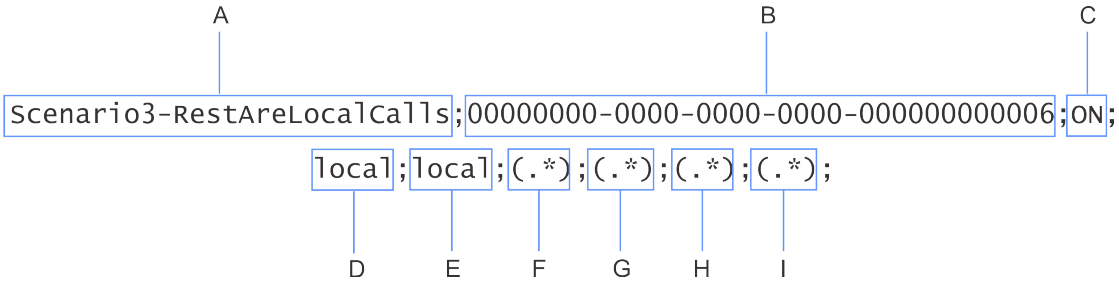
The value labels identified below correspond to the column labels which appear on the *Dial plans* page of the **Sipelia** plugin role.

Value letter	Value label	Description
A	**Name**	The name indicates that this rule will route to the SIP trunk calls made with specific SIP extensions.
B	**Schedule**	The schedule is set to *Always*, meaning that the rule will always be verified.

Value letter	Value label	Description
C	**Status**	The status is set *On*, meaning that the rule is currently active.
D	**Direction from**	The field is set to *local* to look for calls that will be originating from Sipelia Server.
E	**Direction to**	The field is set to *TrunkSIPServerA* to route the calls to that SIP trunk.
F	**Source**	The regular expression is set to **(.*)**, meaning that the call can be made from *any* SIP extension registered to Sipelia Server.
G	**Destination**	The regular expression is set to **\b0*4([0-4][0-9]{2}|500)\b**, meaning the Sipelia Server must to look for destination SIP extensions between 4000 and 4500, as described below: • **\b:** SIP extension boundary. Used together with the same **\b** element at the end of the expression, this requires the whole SIP extension to be matched. Starting or ending characters will not be omitted. • **0*:** Look for any number of zeros before the next character, which is **4**. • **4:** Look for a SIP extension that begins with **4**. • **([0-4]:** Look for a single occurrence of digits **0** through **4** after the first **4**. • **[0-9]{2}:** Look for 2 following occurrences of digits **0** through **9**, which now covers extension 4000 to 4499. • **|500):** Look specifically for **500** to add 4500 in the search pattern. • **\b:** Match a SIP extension boundary.
H	**New source**	The regular expression **(.*)** is used. This means that your extension (the source caller) on Sipelia Server remains unchanged.
I	**New destination**	The regular expression **(.*)** is used. This means that the called extension remains unchanged.

Rule 2: Other calls routed locally

The second rule listed in the dial plan and shown below tells Sipelia Server to route to *TrunkSIPServerA* any other call that does not match the first rule.



Value letter	Value label	Description
A	**Name**	The name indicates that this rule will keep local any other call.
B	**Schedule**	The schedule is set to *Always*, meaning that the rule will always be verified.
C	**Status**	The status is set *On*, meaning that the rule is currently active.
D	**Direction from**	The field is set to *local* to look for calls that will be originating from Sipelia Server.
E	**Direction to**	The field is set to *local* to keep the calls local to Sipelia Server.
F	**Source**	The regular expression is set to **(.*)**, meaning that the call can be made from *any* SIP extension registered to Sipelia Server.
G	**Destination**	The regular expression is set to **(.*)**, meaning that the call can be made to reach *any* SIP extension.
H	**New source**	The regular expression **(.*)** is used. This means that your extension (the source caller) on Sipelia Server remains unchanged.
I	**New destination**	The regular expression **(.*)** is used. This means that the called extension remains unchanged.

Result

Once the dial plan is imported into Config Tool, only the highest priority rule will be apply. This dial plan results in the following:

1. If any SIP extension dials a number between 4000 and 4500, the call will be routed to *TrunkSIPServerA*.
2. If any SIP extension dials any other number, the call will remain local to Sipelia Server.

Dial plan scenario 4: Replacing source SIP extensions

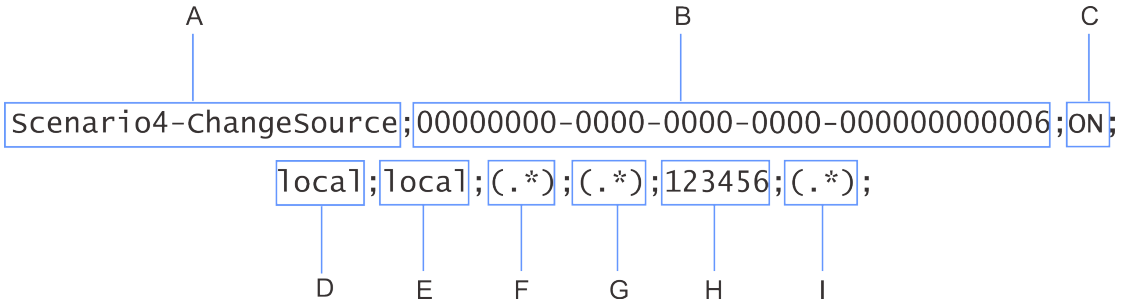
In this scenario, a dial plan rule is used to replace the source SIP extensions with a unique SIP extension that will be displayed to the recipients. This rule can be used to ensure the privacy of the callers.

Scenario

- **What do you want to do?** Display a unique source SIP extension for any call that is made.
- **What you need to do before defining the rule?** Select the SIP extension that will always be used as the source. In this scenario, 123456 is the selected SIP extension.
- **How can you do it?** You need to create a rule that takes any call from *Sipelia™ Server* and replace the source SIP extension with 123456.

Example of the dial plan rule

The following is an example of a dial plan rule that accomplishes the above mentioned scenario.



The value labels identified below correspond to the column labels which appear on the *Dial plans* page of the **Sipelia** plugin role.

Value letter	Value label	Description
A	**Name**	The name of the rule indicates that it will replace the source SIP extension.
B	**Schedule**	The schedule is set to *Always*, meaning that the rule will always be verified.
C	**Status**	The status is set *On*, meaning that the rule is currently active.
D	**Direction from**	The field is set to *local* to look for calls that will be originating from Sipelia™ Server.
E	**Direction to**	The field is set to *local* to keep the calls local to Sipelia Server.
F	**Source**	The regular expression is set to **(.*)**, meaning that the call can be made from *any* SIP extension registered to Sipelia Server.
G	**Destination**	The regular expression is set to **(.*)**, meaning that the call can be made to reach *any* SIP extension.

Value letter	Value label	Description
H	**New source**	The regular expression is set to **123456** to change the SIP extension with this constant value.
I	**New destination**	The regular expression **(.*)** is used. This means that the called extension remains unchanged.

Result

Once this dial plan rule is imported into Config Tool, if any SIP extension registered to Sipelia Server dials any other SIP extension, the source extension 123456 will be displayed to the recipients.

Dial plan scenario 5: Removing prefix on source SIP extensions from a SIP trunk

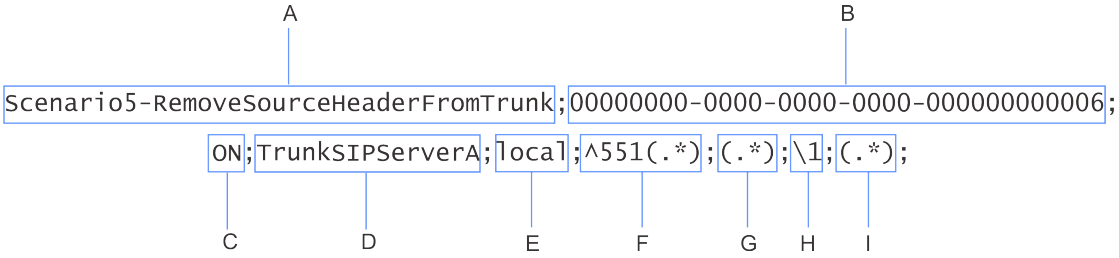
In this scenario, a dial plan rule is used to remove the prefix used in source SIP extensions for calls received from a SIP trunk, so that it is not displayed to the recipients. This scenario can be combined with scenario 1 to have the source extensions displayed with no prefix when received by the recipients registered to *SIP Server A*.

Scenario

- **What do you want to do?** Remove the prefix included in source SIP extensions when calls are received from a SIP trunk.
- **What you need to do before defining the rule?** Add a SIP trunk in Config Tool for *SIP Server A* and name it *TrunkSIPServerA*. This trunk name must be unique, so that the dial plan rule that you will create will not conflict with other rules, and your rule will be applied correctly. You also need to know the prefix used by the SIP trunk for its extensions.
- **How can you do it?** Create a rule that takes any call from a SIP trunk and remove the prefix used in the source SIP extension.

Example of the dial plan rule

The following is an example of a dial plan rule that accomplishes the above mentioned scenario.



The value labels identified below correspond to the column labels which appear on the *Dial plans* page of the **Sipelia** plugin role.

Value letter	Value label	Description
A	**Name**	The name of the rule indicates that it will remove the prefix from source SIP extensions received from the SIP trunk.
B	**Schedule**	The schedule is set to *Always*, meaning that the rule will always be verified.
C	**Status**	The status is set *On*, meaning that the rule is currently active.
D	**Direction from**	The field is set to *TrunkSIPServerA* to look for calls that will be originating from the SIP trunk.
E	**Direction to**	The field is set to *local* to route the calls to Sipelia™ Server.
F	**Source**	The regular expression is set to **^551(.*)** to look for *any* source SIP extension that begins with prefix 551.

Value letter	Value label	Description
G	**Destination**	The regular expression is set to **(.*)**, meaning that the call can be made to reach *any* SIP extension.
H	**New source**	The regular expression is set to **\1**, which indicates that the rule will use the value of the first regular expression group, which was defined in the *Source* value. This means that the prefix will be discarded and the source SIP extension becomes **(.*)**.
I	**New destination**	The regular expression **(.*)** is used. This means that the called extension remains unchanged.

Result

Once this dial plan rule is imported into Config Tool, any source SIP extension with prefix 551 received from *TrunkSIPServerA* will be displayed without the prefix to the recipients. For example, if a call received from the SIP trunk is made from extension 5513005, the recipient registered to Sipelia Server will only see 3005.

Dial plan scenario 6: Forwarding calls to another SIP extension on schedule

In this scenario, dial plan rules are used to define a range of SIP extensions for calls that must always reach the recipients, while other SIP extensions are automatically forwarded to a new destination. This can be useful to route calls from SIP intercoms to a call center during the night for example.

Scenario

- **What do you want to do?** During off-hours for example, calls made to SIP extensions between 4000 and 4500 must be routed normally to the requested recipients, while other calls must be forwarded to extension 1001, which can be the extension of a SIP endpoint (or SIP client) in a security office for example.
- **What you need to do before defining the rule?** Define the range of SIP extensions that must always be routed and the specific SIP extension to which other calls will be forwarded. You also need to have a schedule defined in Security Center to be used in the dial plan rule.
- **How can you do it?** You need to define two separate rules listed in the right order in your dial plan:
  - The first rule looks for any call made to reach SIP extensions between 4000 to 4500, and route them normally.
  - The second rule will take calls to any other SIP extension and forward them to extension 1001.

IMPORTANT: The rules need to be in the right order in the dial plan to work as described in this scenario.

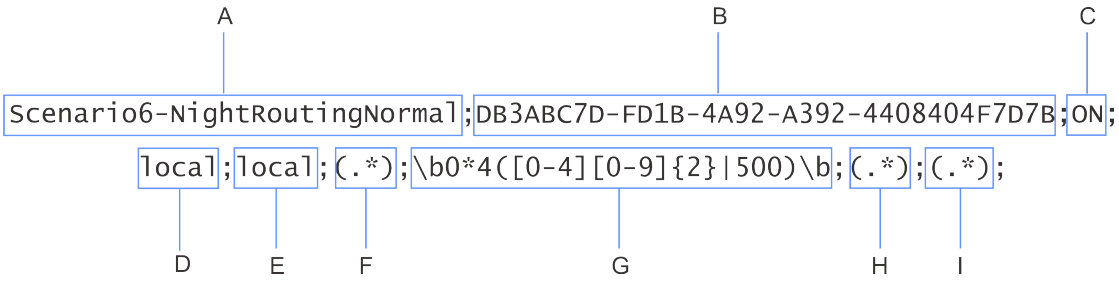
Example of the dial plan

The following is an example of a dial plan that accomplishes the above mentioned scenario.

1. Scenario6-NightRoutingNormal;DB3ABC7D-FD1B-4A92-A392-4408404F7D7B;ON;local;local;(.*);\b0*4([0-4][0-9]{2}|500)\b;(.*);(.*);
2. Scenario6-NightRoutingSpecial;DB3ABC7D-FD1B-4A92-A392-4408404F7D7B;ON;local;local;(.*);(.*);(.*);1001;

Rule 1: Normal night routing

The first rule listed in the dial plan shown below tells Sipelia™ Server to look for destination SIP extensions between 4000 and 4500, and route them normally.



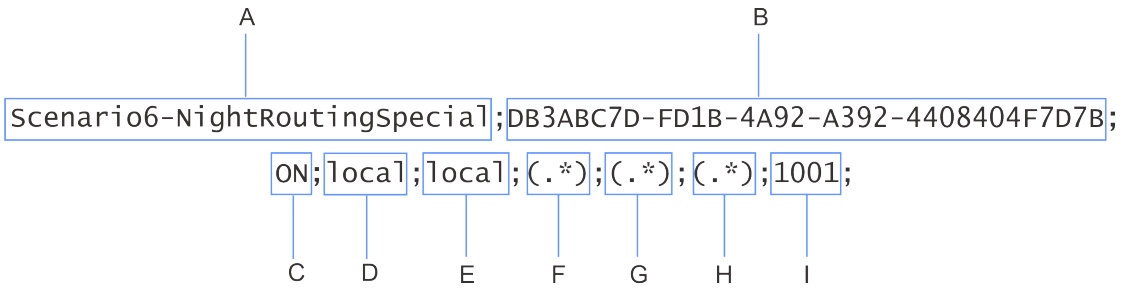
The value labels identified below correspond to the column labels which appear on the *Dial plans* page of the **Sipelia** plugin role.

Value letter	Value label	Description
A	**Name**	The name of the rule indicates that the calls are routed normally.

Value letter	Value label	Description
B	**Schedule**	The schedule is set to *Off-hours*, meaning that the rule will be applied only during that time.
C	**Status**	The status is set *On*, meaning that the rule is currently active.
D	**Direction from**	The field is set to *local* to look for calls that will be originating from Sipelia Server.
E	**Direction to**	The field is set to *local* because the calls will remain local to Sipelia Server when the rule is applied.
F	**Source**	The regular expression is set to **(.*)**, meaning that the call can be made from *any* SIP extension.
G	**Destination**	The regular expression is set to **\b0*4([0-4][0-9]{2}|500)\b**, meaning that Sipelia Server must look for called SIP extensions between 4000 and 4500, as described below: • **\b:** SIP extension boundary. Used together with the same **\b** element at the end of the expression, this requires the whole SIP extension to be matched. Starting or ending characters will not be omitted. • **0*:** Look for any number of zeros before the next character, which is **4**. • **4:** Look for a SIP extension that begins with **4**. • **([0-4]:** Look for a single occurrence of digits **0** through **4** after the first **4**. • **[0-9]{2}:** Look for 2 following occurrences of digits **0** through **9**, which now covers extension 4000 to 4499. • **|500):** Look specifically for **500** to add 4500 in the search pattern. • **\b:** Match a SIP extension boundary.
H	**New source**	The regular expression **(.*)** is used. This means that your extension (the source caller) on Sipelia Server remains unchanged.
I	**New destination**	The regular expression **(.*)** is used. This means that the called extension remains unchanged.

Rule 2: Special night routing

The second rule listed in the dial plan and shown below tells Sipelia Server to look for any call that does not match the first rule, and forward it to extension 1001.



The value labels identified below correspond to the column labels which appear on the *Dial plans* page of the **Sipelia** plugin role.

Value letter	Value label	Description
A	**Name**	The name of the rule indicates that the calls are following a special route.
B	**Schedule**	The schedule is set to *Off-hours*, meaning that the rule will be applied only during time.
C	**Status**	The status is set *On*, meaning that the rule is currently active.
D	**Direction from**	The field is set to *local* to look for calls that will be originating from Sipelia Server.
E	**Direction to**	The field is set to *local* because the calls will remain local to Sipelia Server when the rule is applied.
F	**Source**	The regular expression is set to **(.*)**, meaning that the call can be made from *any* SIP extension.
G	**Destination**	The regular expression is set to **(.*)**, meaning that the call can be made to reach *any* SIP extension.
H	**New source**	The regular expression **(.*)** is used. This means that your extension (the source caller) on Sipelia Server remains unchanged.
I	**New destination**	The regular expression **1001** is used. This means that the call will always be forwarded to SIP extension 1001.

Result

Once the dial plan is imported into Config Tool, only the highest priority rule will apply. This dial plan results in the following:

1. When the schedule is satisfied, if any SIP extension dials a number between 4000 and 4500, the call will remain local to Sipelia Server and will be routed normally to the requested recipient.
2. When the schedule is satisfied, if any SIP extension dials any other number, the call will automatically be forwarded to SIP extension 1001.

About importing dial plans

Dial plan rules are stored in comma-separated values (CSV) text files, and these files can contain multiple rules.

Importing dial plans

Once dial plan rules have been defined in the .csv file and a SIP trunk has been added, you must import the dial plan files into Config Tool so that the rules can be applied accordingly.

Before you begin

- Add a SIP trunk.
- Define applicable dial plan rules.

What you should know

If you delete ( ) a dial plan rule that you have imported, the rule is no longer available to the dial plan system of Sipelia™ Server. If you want the deleted rule to be available again, you must reimport the dial plan file which contains the rule. If you want the dial plan system to no longer apply a rule, simply make the rule inactive, and keep it as part of your imported rules in case you ever want to reapply the rule again.

When updating a dial plan rule that has already been imported once, you do not need to delete the existing rule from the *Dial plans* page. In the dial plan file, simply make the necessary changes to the rule, make sure not to change the name of the rule, and then reimport the rule. Upon import, dial plan rules with new names get added to the list of rules; those whose names are already part of the list, are simply updated.

Procedure

1 From the Config Tool homepage, open the *Plugins* task.

2 Select the **Sipelia**™ plugin role, and then click **Dial plans**.

3 Click **Import dial plan rules from file** ( ).

4 Select the dial plan file, and then click **OK**.

The dial plan is imported, and the rules contained within your file appear on separate lines on the *Dial plans* page.

IMPORTANT: If a dial plan rule does not have the correct number of comma-separated values, the rule is not imported and an error is generated.

5 If your dial plan file contains issues, click the **Show errors and warnings** button to open the *Parser result* window.

This window classifies the issues by type (error or warning), and provides information on each issue. Warnings do not need to be corrected, but errors do. Simply modify the dial plan file accordingly, and then reimport the file.

6 For each dial plan rule that you import, you can edit the following:

- **Schedule:** The Security Center schedule that defines when this rule must be in effect. The rule is applied only if the schedule condition is met.
- **Status:** The status of the rule. Only *Active* rules are applied.

7 (Optional) To change the order in which the dial plan rules are applied, use the arrow buttons to move the rules up or down.

8 Click **Apply**.

5

Troubleshooting

This section includes the following topics:

- "Unable to establish communication with the Sipelia Server" on page 91
- "Message broker connection failed to all the configured hosts" on page 92
- "Cannot add SIP intercom devices" on page 93
- "Cannot see the Sipelia icon in the notification tray" on page 94
- "Security Desk cannot connect to Sipelia Server" on page 95
- "Sipelia plugin role failed to load" on page 96
- "Cannot register to Sipelia Server from Security Desk" on page 97
- "Cannot make calls between two SIP endpoints" on page 98
- "Sipelia calls from Security Desk are delayed or are not delivered" on page 99
- "No video or distorted video displayed during Sipelia calls" on page 100
- "Audio and video aren't being recorded during Sipelia calls" on page 101
- "No audio, or distorted audio in Sipelia calls" on page 103
- "DTMF tones not working in Sipelia" on page 105
- "Enabling file logging for debug traces on the Sipelia Server" on page 106
- "Waiting for Sipelia to be initialized" on page 107
- "Unable to add a new intercom entity because licenses have been exceeded " on page 108
- "Call Report task does not replay the selected session" on page 109

Unable to establish communication with the Sipelia Server

When adding a SIP intercom in Config Tool, the error message **Unable to establish communication with the server** is displayed.

What you should know

This error typically occurs when there is a connection issue with the configuration service between Config Tool and Sipelia™ Server.

Procedure

1 From the Config Tool homepage, open the *Plugins* task.

2 Select the Sipelia plugin role, and then click **General**.

3 Make sure that **Configuration service port** is set to the proper value.

4 Make sure that this port is not being blocked anywhere on the network.

Message broker connection failed to all the configured hosts

The Sipelia™ plugin role displays the error message **Message broker connection failed to all the configured hosts:** and the connection to Sipelia Server is not possible from Security Desk.

Before you begin

- Ensure that the Message broker server parameters are correct.
  - The server address must be reachable from the machine where Sipelia Server and Sipelia Client are installed.

What you should know

- This error typically occurs when your Sipelia Message broker configuration is incorrect, or your Message broker server is not reachable from the machine hosting the Sipelia system.

Procedure

1 Restart the Sipelia plugin role. If the problem is still present, continue with the next step.

2 Ensure that your Message broker configuration is correct. Verify the username, password, address, and port.

Cannot add SIP intercom devices

When adding a SIP intercom in Config Tool, the error message **The number of Sipelia licenses (standard or advanced) has been exceeded.** is displayed.

What you should know

This error typically occurs when you do not have enough standard and/or advanced licenses in your system to support the number of SIP intercom devices that you want to add.

- Remove one or more SIP intercom devices from your system.
- Increase the number of licenses.

Cannot see the Sipelia icon in the notification tray

A user cannot see the Sipelia™ icon in the notification tray of Security Desk.

What you should know

This issue typically occurs when Sipelia Client has not been installed properly or has been installed with an unsupported Security Center version. For more information on Security Center compatibility with Sipelia, see the *Sipelia™ Release Notes*.

Procedure

1 Ensure that you are using a compatible version of Security Center.

2 If the problem is still present, restart Security Desk.

3 If the problem is still present, restart the computer on which Security Desk is running.

4 If you are still unable to see the icon in the notification tray, uninstall and re-install Sipelia Client on the computer.

Security Desk cannot connect to Sipelia Server

If Security Desk cannot connect to the Sipelia™ Server, verify that the IP address and port properties are properly set.

What you should know

Connection issues typically occur when IP addresses or ports are not configured properly, or when the network is blocking packets from being exchanged between two endpoints.

Procedure

1 From the Config Tool homepage, open the *Plugins* task.

2 Select the Sipelia plugin role, and verify that the role is up and running.

3 Click **General**, and make sure that the address and port properties are set to the proper values.

4 Click **Servers**, and make sure that **SIP port** is set to the proper value.

5 Make sure that the ports are not being blocked anywhere on the network.

6 Open the *Network* task, and then select the server that is hosting the Sipelia plugin role.

7 Click **Properties**, and make sure that the first IP address shown in **Private addresses**, that is, the IP address listed at the top, is the one that you want to be used by the server.

Sipelia Server automatically uses the first IP address shown in the list.

8 Log on to Security Center with Security Desk.

9 Click **Options** > **Sipelia** > **Advanced**, and make sure that **UDP port range** is set properly.

10 On the Sipelia Server, browse to folder *C:\ProgramData\Genetec Sipelia\SipServer* and open *SipServer.config*.

11 Make sure that *MinimumPortRange* and *MaximumPortRange* values are set properly in the file.

Sipelia plugin role failed to load

If you are running more than one instance of the Sipelia™ plugin, the system generates an error and the role fails to load.

What you should know

Multiple instances of the Sipelia plugin on one system are not supported. If you have more than one instance of the plugin, the additional instances will remain in error state with a message: *Role failed to load - Only one instance of the Sipelia Plugin can be running in the system*.

Procedure

1 In Config Tool, open the *Plugins* task.

2 Decide which instance of the Sipelia plugin you want to keep.

3 Delete all other instances of the Sipelia plugin.

Cannot register to Sipelia Server from Security Desk

If users can't register to Sipelia™ Server from Security Desk, you should verify that the VoIP properties are properly set for those users.

What you should know

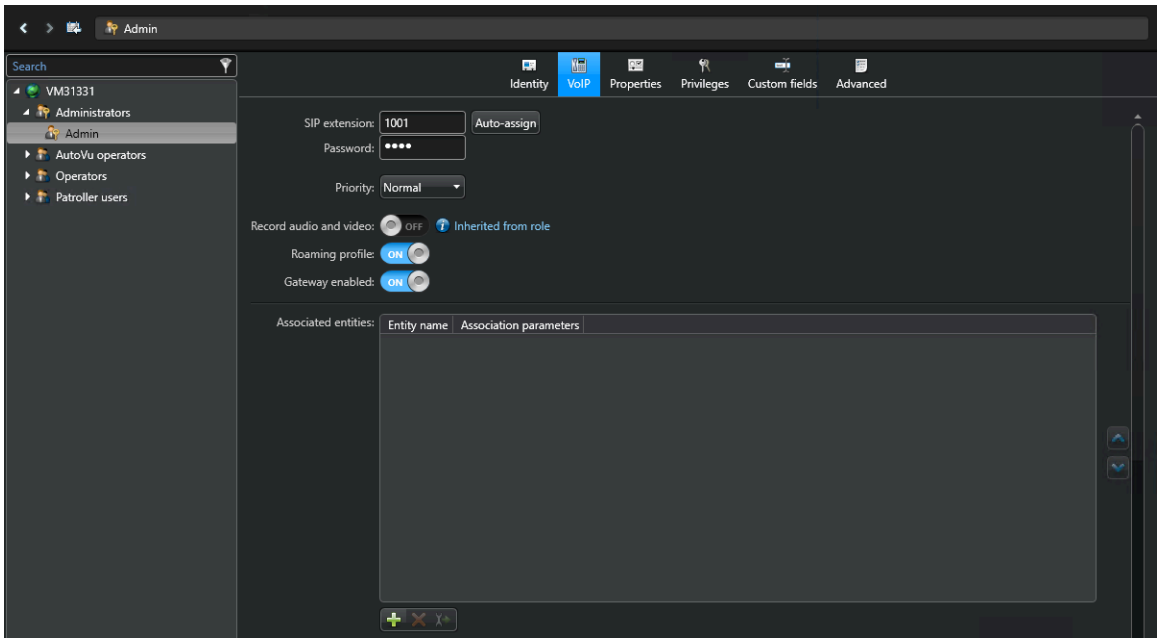
Registration issues typically occur when Security Desk can connect to the Sipelia Server but SIP extensions or passwords aren't configured properly.

Procedure

1 Log on to Security Center with Config Tool.

2 Open the *Security* task, and then select the user entity involved in the connection issue.

3 In the *VoIP* page, make sure that the **SIP extension** and **Password** properties have been configured with the proper values.



Cannot make calls between two SIP endpoints

If users cannot make or receive Sipelia™ calls with Security Desk, the messages might be blocked somewhere on the network.

What you should know

When users are registered to Sipelia Server but cannot make or receive calls, this typically occurs when the network is blocking packets from being exchanged between two SIP endpoints.

Procedure

1. Log on to Security Center with Security Desk.
2. Using a network protocol analyzer (Wireshark, for example), try to make a call and verify that SIP messages are exchanged between the two SIP endpoints.
3. If SIP packets are being exchanged, verify that SDP and RTP packets are also exchanged.
4. Make sure that the IP addresses and port numbers used in the SIP and SDP packets are correct.
5. If you can see SIP, SDP, and RTP packets exchanged on the network between the two SIP endpoints, but users are still not able to make or receive calls, try to restart the Genetec™ Server.

Sipelia calls from Security Desk are delayed or are not delivered

If calls are delayed or not delivered from Security Desk to another SIP client or endpoint, you might have to disable *NAT Traversal mode* on the Sipelia™ Server.

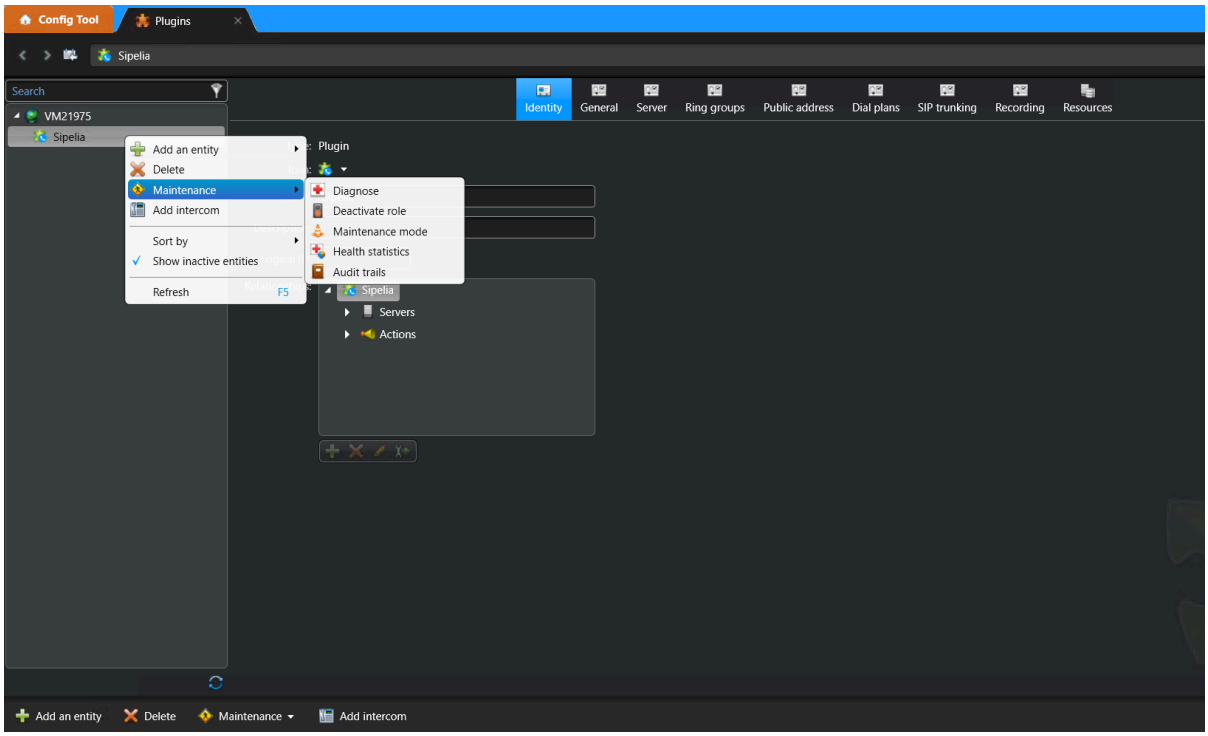
What you should know

This issue typically occurs when:

- Calls are made from Security Desk to another SIP client or endpoint.
- Security Desk is inside a different subnet or different network than the Sipelia Plugin.
- Systems with the Sipelia Plugin do not have access to the Internet.

Procedure

1 Deactivate the Sipelia Plugin role.
In the *Plugins* task, right-click the Sipelia Plugin and select **Maintenance** > **Deactivate role**.



2 Open the Sipelia Server configuration file (*C:\ProgramData\Genetec Sipelia\SipServer\SipServer.config*).

3 Change the value of the `NatTraversalMethod` property from *Default* to *None*.

4 Activate the Sipelia plugin role.
In the *Plugins* task, right-click the Sipelia Plugin and select **Maintenance** > **Activate role**.

No video or distorted video displayed during Sipelia calls

If you cannot see video during SIP video calls, or if the video is distorted, there might be a video profile incompatibility between the SIP endpoints participating in the communication.

What you should know

Sipelia™ Server performs the media transcoding if both parties use different codecs to stream their audio and video. However, even if a common codec is in use, an incompatibility between the video profiles of both parties might occur. A video profile is a set of parameters that represents the video stream that is sent from a given communication participant.

Procedure

1 If your intercom unit uses the H.263 or H.263+ (1999) codecs and if Sipelia uses the .H264 codec, try to reconfigure the intercom unit to also use the H.264 codec.

NOTE: The highest matching preference is always chosen. For example, if both endpoints support H.264 and H.263, the connection is established using H.264. You can see the Security Desk video codec settings in **Options** > **Sipelia** in the *Advanced* section.

2 If the problem is not resolved, try to use different video parameters on your intercom unit, for example, resolution, bitrate, and the frames per second ratio.

3 If the problem is still not resolved, use a network protocol analyzer (for example, Wireshark) to verify that RTP (UDP) packets are exchanged between the two SIP endpoints.

- **UDP port range:** This is the port range for the User Diagram Protocol (UDP). The UDP ports are used by SIP clients to send and receive communication data. The default range for the Sipelia plugin and Security Desk is from 20000 to 20500. It is recommended that you keep the default settings. You should only change these settings if your network environment requires other values.

Audio and video aren’t being recorded during Sipelia calls

If you can see video during SIP video calls, but the recording of the call isn’t in the database, you might have recording options that aren’t configured properly.

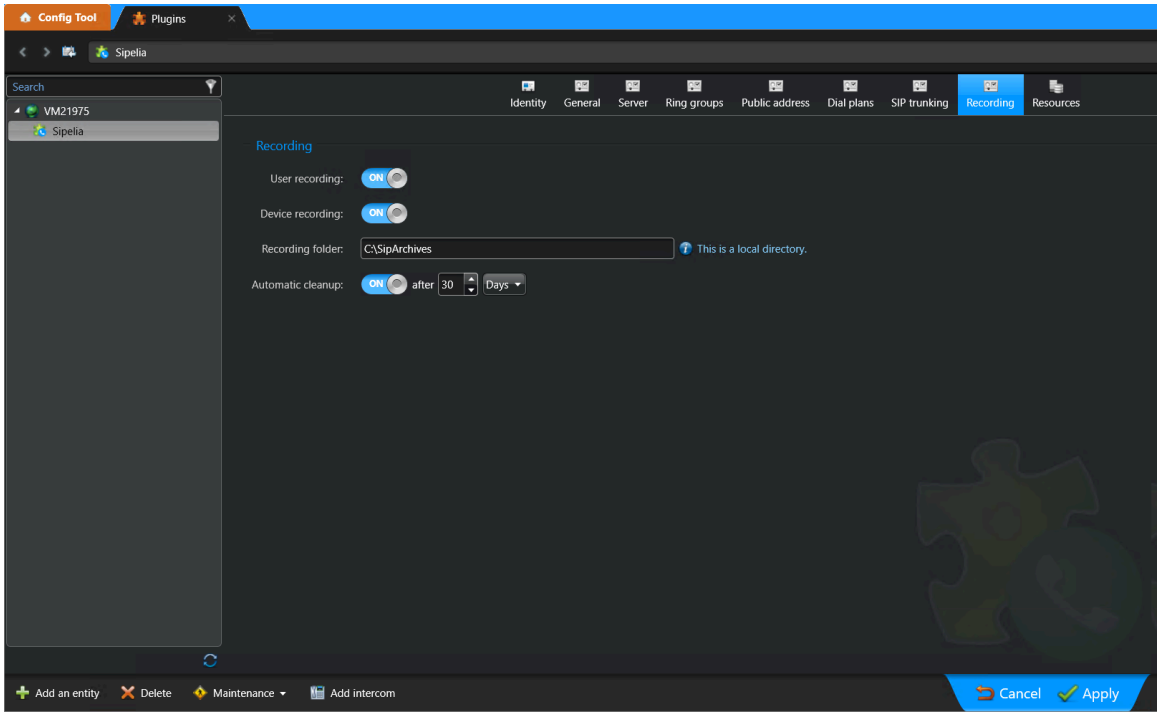
Procedure

1 Verify that your license supports the recording of call sessions.

2 From the Config Tool homepage, open the *Plugins* task.

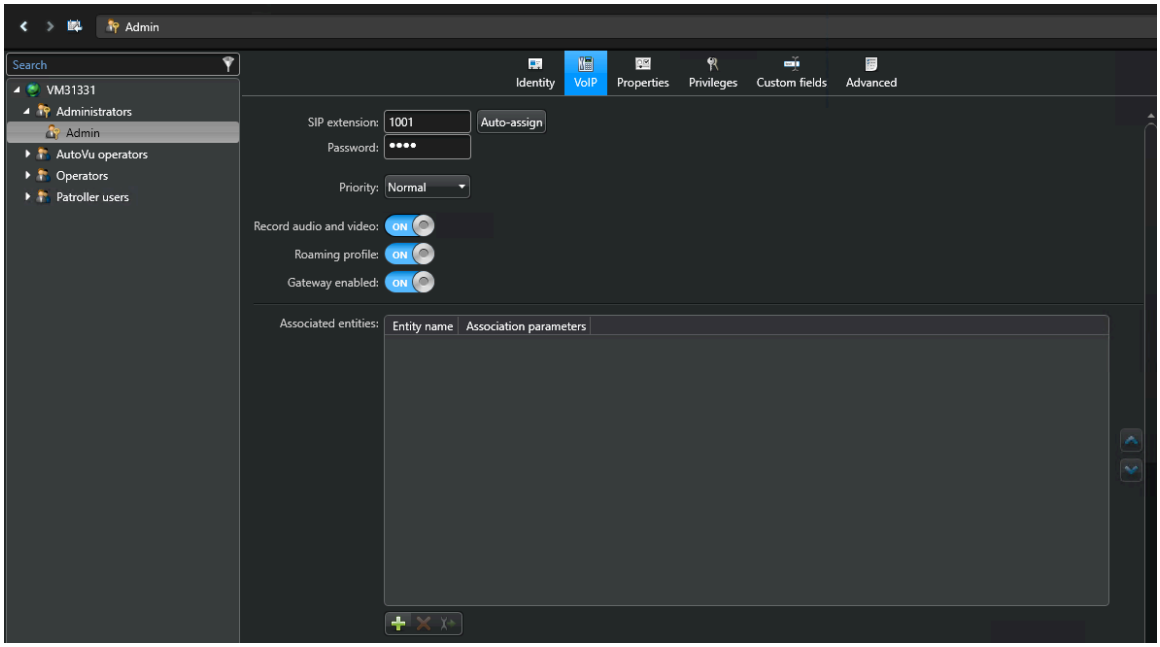
3 Select the Sipelia™ plugin role, and then click **Recording**.

4 Make sure that **User recording** and **Device recording** are enabled.



5 If you have no recording associated with specific users, open the *Security* task, and then select the user entity involved in the issue.

6 In the *VoIP* page, make sure that the **Record audio and video** property is turned on or inherits the default value from the Sipelia plugin role.



7 If you still have problems recording call sessions, try restarting the Genetec™ Server.

No audio, or distorted audio in Sipelia calls

If there is no audio, or if you are experiencing poor audio quality when making Sipelia™ calls, it might be possible to improve audio quality by using a different audio codec, and by verifying the RTP packet size defined for the SIP intercoms.

Before you begin

- Ensure that the USB headset you are using is configured and is working properly.
- Create a backup of the following file:

  C:\ProgramData\Genetec Sipelia\SipServer\SipServer.config

What you should know

- Enabling and disabling video and audio codecs is done through an XML configuration file. In this file, you can find the following XML elements: **AudioCodecs** and **VideoCodecs**. The **Codec** element within them contains the **Enabled** attribute that defines the state of each codec.
- Audio quality in Sipelia depends on the audio codec that you selected. If required, you can enable high-quality audio.
- In cases where both sides of the communication use a different codec, Sipelia Server transcodes the media stream to ensure that communication is still possible.

No audio:

- Intercoms can support one or multiple audio codecs. The most commonly used audio codecs are PCMA, PCMU, and G722. Research the audio quality and network usage associated with the supported codecs.
- Sipelia supports several audio codecs. SIP intercoms must support at least one of the codecs that are supported by Sipelia.
- If changing the codecs on the Sipelia Server does not resolve the issue, you might need to configure the audio codecs used by the SIP intercoms. To investigate this, refer to the configuration page of each SIP intercom. You can also run a Wireshark capture to verify which codecs each device is using.

Distorted audio:

- All systems that you use with Sipelia (intercoms, servers, hardware, and software) should use an RTP packet size if 20 ms. Refer the documentation provided with the intercom hardware to see if the RTP packet size is configurable.
- In Security Desk the noise and echo cancellation is turned on by default. With some particular environment and intercom units, the final call audio level is too low to be played properly by the intercom speaker. In that case you can turn off the noise and audio cancellation by editing the UserOptions.xml configuration file. To configure the XML file:
  - Locate the `IsNoiseAndEchoCancellationEnabled` XML element (in *%PROGRAMDATA%\Genetec Sipelia* folder) and change the value from `True` to `False`.
  - The following code shows how to use this feature.

```
<UserOptions>
  <Microphone></Microphone>
  <Speaker></Speaker>
  <Camera></Camera>
  <CameraQualityProfile>High</CameraQualityProfile>
  <Microphone><Microphone>
  <Microphone><Microphone>
  ...
  <IsNoiseAndEchoCancellationEnabled>True</IsNoiseAndEchoCancellationEnabled>
</UserOptions>
```

  - Restart Security Desk to apply the change.

- The distorted audio maybe also be caused by particular audio codec. Reconfigure the system to use a different audio codec, then test the system and evaluate the audio quality. If the audio quality is poor, enable a different audio codec and perform another audio quality test.

Procedure

1 Enable or disable codecs as required.

2 Test the system and evaluate the audio quality. If the audio quality is poor, enable a different audio codec and perform another audio quality test.

DTMF tones not working in Sipelia

If pressing a number button on the Security Desk keypad does not produce an audible touch-tone signal (DTMF) at the SIP intercom, verify that the **telephone-event** audio codec is enabled on the Sipelia™ Server, and that an audio channel is established between the Sipelia Server and the SIP intercom.

Before you begin

- Create a backup of the following file:

  C:\ProgramData\Genetec Sipelia\SipServer\SipServer.config

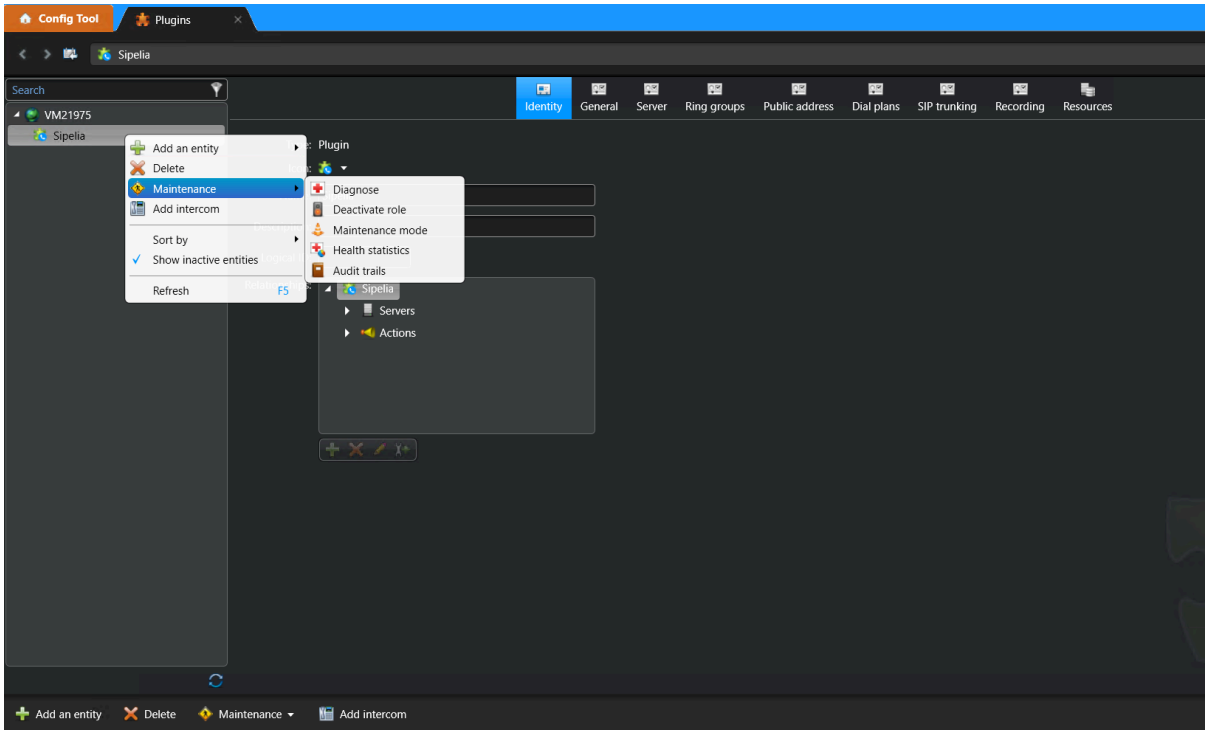
What you should know

- DTMF tones in Sipelia are transported by the RTP protocol as the RTP EVENT payload (RFC 2833).
- DTMF tones do not depend on the type of audio codec which was used to enable the audio channel between the call participants. For more information on configuring which codecs are used for audio communication, refer to the related audio troubleshooting information.

Procedure

1 Deactivate the Sipelia Plugin role.

In the *Plugins* task, right-click the Sipelia Plugin and select **Maintenance** > **Deactivate role**.



2 Open the following configuration file:

C:\ProgramData\Genetec Sipelia\SipServer\SipServer.config

3 In the *AudioCodecs* section of the configuration file, note whether the **Enabled** attribute is `True` or `False` for each of the audio codecs.

4 To enable DTMF, set the **telephone-event** codec to `True`.

5 Save the file and reactivate the Sipelia plugin role.

Enabling file logging for debug traces on the Sipelia Server

File logging for debug traces can be enabled on the Sipelia™ Server side, the Sipelia Client side, or both. The nature of the problem to be investigated will determine if and where the debugging traces need to be enabled.

Before you begin

You need administrator privileges to be able to modify the required configuration files.

Procedure

To enable file logging for debug traces on the Server side:

1 Stop the Sipelia Server.

2 Go to the configuration folder for the server logs.
It is located under the Sipelia program data folder as shown in the following example:

C:\ProgramData\Genetec Sipelia\Logs\Server

3 Open the file *NLog.dll.nLog*.

4 On line 12, replace text A with text B:

Text A: `logger name="*" levels="Info,Warn" writeTo="RegularLogFile"`

Text B: `logger name="*" levels="Trace,Debug,Info,Warn" writeTo="RegularLogFile"`

5 Save the file and restart the Sipelia Server.

To enable file logging for debug traces on the Client side:

1 Go to the configuration folder for the client logs.
It is located under the Sipelia program data folder as shown in the following example:

C:\ProgramData\Genetec Sipelia\Logs\Client

2 Open the file *NLog.dll.nLog*.

3 On line 12, replace text A with text B:

Text A: `logger name="*" levels="Info,Warn" writeTo="RegularLogFile"`

Text B: `logger name="*" levels="Trace,Debug,Info,Warn" writeTo="RegularLogFile"`

4 Save the file.

The system will now log debugging traces in log files. The output files can be found in the following directory:

C:\ProgramData\Genetec Sipelia\Logs

IMPORTANT: To avoid filling the hard disk with unnecessary information, it is recommended to revert the changes to the original configuration as soon as the required investigation has been completed.

Waiting for Sipelia to be initialized

After installing the Sipelia™ plugin, the plugin role is in a warning state (yellow) with the error messages *Waiting for Sipelia to be initialized* and *Message broker connection pending*.

What you should know

You might notice the following issues:

- The Sipelia plugin is always in a warning state (plugin icon is yellow).
- You receive the error messages *Waiting for Sipelia to be initialized* and *Message broker connection pending*.
- In the **System > Roles**, the tabs for the Sipelia plugin are not active.
- The Sipelia Server log stops after the message *Received a new IP: A.B.C.D*.
- This error is caused by a network issue where the software is unable to start because the received IP address is not operational on the network to which it belongs. This can happen when the workstation on which the Sipelia plugin is running has more than one network card, but not all cards are communicating with the rest of the system.
- To resolve this issue, ensure that the Security Center system is using the correct network card. This configuration is done on the Security Center *Server Admin* web page. The Sipelia role will only use the network cards that are selected in Server Admin.

Unable to add a new intercom entity because licenses have been exceeded

If you are unable to add a new intercom entity because you exceeded the number of installed licenses, and the Security Desk and Config Tool license information shows that the license is still available, you need to synchronize the license count between the Security Center Directory and the Sipelia™ plugin.

What you should know

This issue typically occurs when you frequently add and remove intercoms (or their extension) in your system. For each assigned extension, one license is used.

Procedure

1 Make sure that you do not exceed the number of installed standard and advanced licenses.

NOTE: If the standard and advanced license counts do not match, the system will choose the lower count for both types of licenses. For more information, see How Licensing Works in Sipelia.

2 Restart the Sipelia plugin.

3 Restart the Security Center applications (Security Desk and Config Tool).

Call Report task does not replay the selected session

When trying to replay a Sipelia™ session in the Security Desk *Call report* task, the selected session does not replay properly.

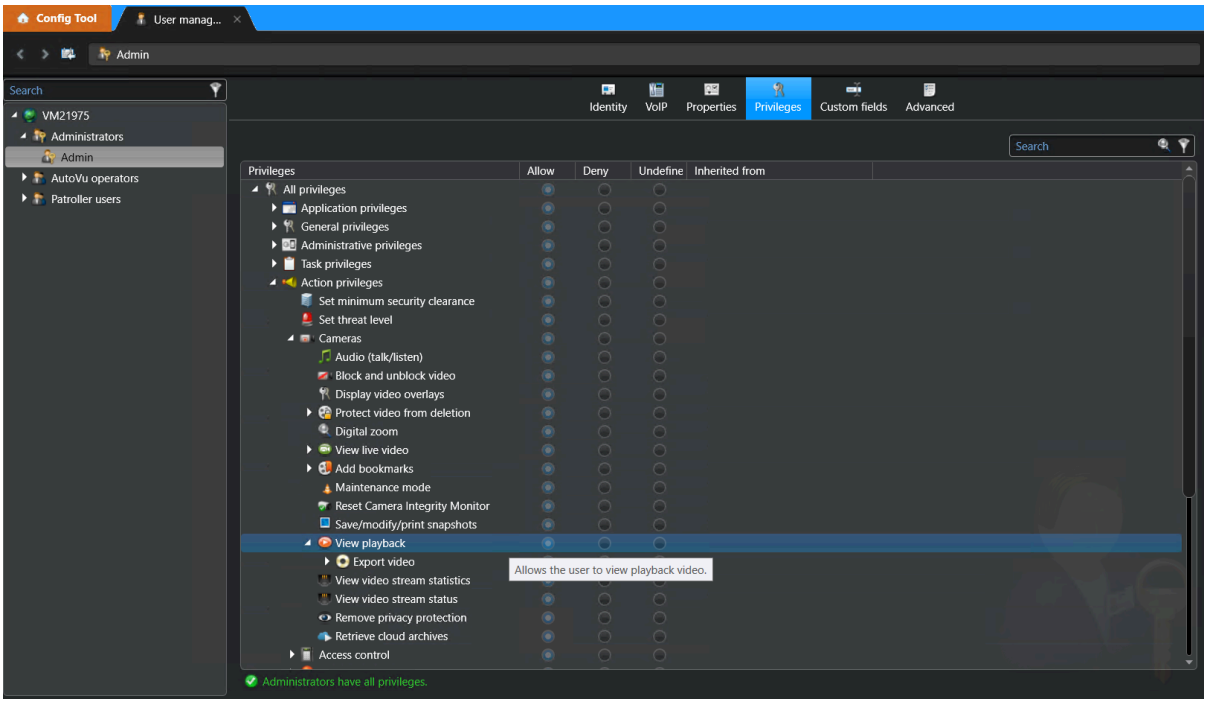
Before you begin

- Ensure that you have the correct privileges to view recorded video in Security Center.
- It is recommended to use a physical machine to run the Security Desk *Call report* task.

What you should know

The *Call report* task is a service that allows users to replay and investigate archived calls. The Sipelia archive mechanism combines different kinds of events from the client side and the server (call status changed, recording started, door unlocked, DTMF tone sent) and stores them accordingly with the audio and video recordings. The recordings include the audio and video stream from the peripherals of call participants as well as the Security Center Archiver recordings from the camera entities that were attached to call participants.

- In the *Security* task, **Privileges** page, make sure that the **View playback** privilege is allowed.



- Check if the sound device is present and configured properly in your operating system.
- If a particular session is not present in the report and should be according to the filters, or if session events are out of order, be sure that the time is synchronized between the machine that hosts the Sipelia plugin role and the machine that hosts Security Desk in that session.
- If you are connecting by a Remote Desktop or Hyper-V Virtual Machine Connection, to access the machine on which Call Report is used, try another physical machine.
- If the video from associated camera entities is not replayed, check if the Archiver role is running without any warnings or errors. If the Archiver role state is correct, check if a given camera recording on a specific time frame is present inside the Archiver. It's possible that during the call time, the Archive role was offline.
- If the audio or video from call participants does not replay, check if the particular recording files exist in the Sipelia recording folder: **Config Tool** > **Sipelia™ role** > **Recording tab**. If the recording files do not

exist, then during the call no audio and video stream was recognized by Sipelia Plugin. Check the audio/video peripherals of the call participants. Confirm, on the machine that hosts the Sipelia plugin role, if audio and video RTP stream is properly received by the Sipeliaserver.

- *Call report* task downloads the recording files by TCP connection on the port number configurable in **Config Tool** > **Sipelia™ role** > **General tab** > **Session Transfer port**. Check if there is a network connectivity problem between the client that runs Report Task and the machine that hosts the Sipelia plugin role. Also verify if Sipelia Plugin can access the configured Sipelia recording folder if it is located on a network share storage.

Glossary

call dialog box

The call dialog box is a Sipelia-related dialog box that appears from the notification tray within Security Desk once an incoming call comes in from a SIP endpoint. In the call dialog box, users can manage their calls and their list of contacts and favorites, and set their availability status.

callee

A callee is an entity or user agent that receives SIP call requests from a caller.

Call report

The *Call report* task is an investigation task. You can use it to review call sessions and generate reports. Among the possible actions that can be performed in this task, users can view the call logs for all sessions, watch playback video of all recorded call sessions, and see the bookmarks that have been added to video sequences of associated cameras.

call session

A call session is the sequence of events or activities that occur from the time a SIP call is started to the time that it ends, including all call transfers. For example, if a call is transferred twice, the sequence of events that occur in both those transfers are all part of the same call session. In a Security Center system equipped with the Sipelia™ module, call sessions can be reviewed and exported through the *Call report* task within Security Desk.

Config Tool

Config Tool is the Security Center administrative application used to manage all Security Center users and to configure all Security Center entities such as areas, cameras, doors, schedules, cardholders, patrol vehicles, ALPR units, and hardware devices.

conversation window

The conversation window is a Sipelia™-related window that opens within Security Desk once a SIP call has been accepted by either the caller or the recipient of a call. From the conversation window, Security Center users can manage conversations, forward calls, and view associated video when available.

dial plan

A dial plan is a collection of rules that defines how calls are routed locally or between two SIP trunks. Dial plans ensure that calls are routed and rerouted correctly, and they also allow administrators to block calls to certain geographic locations or ensure the privacy of the callers.

entity

An entity represents anything in your system that requires configuration. This can be a physical device, such as a camera or a door, or an abstract concept, such as an alarm, a schedule, a user, a role, a plugin, or an add-on.

Genetec™ Server

Genetec™ Server is the Windows service that is at the core of Security Center architecture, and that must be installed on every computer that is part of the Security Center's pool of servers. Every such server is a generic computing resource capable of taking on any role (set of functions) you assign to it.

network address translation

Network address translation is the process of modifying network address information in datagram (IP) packet headers while in transit across a traffic routing device, for the purpose of remapping one IP address space into another.

notification tray

The notification tray contains icons that allow quick access to certain system features, and also displays indicators for system events and status information. The notification tray display settings are saved as part of your user profile and apply to both Security Desk and Config Tool.

record type

In Security Center, a *record type* defines the data format and display properties of a set of records that you can share across the entire system through the Record Fusion Service role.

regular expression

A regular expression is a sequence of symbols used by a regular expression engine to identify all the strings of characters that match a specific search pattern without having to list all the possible discrete values that must be returned. The Microsoft's .NET Framework Regular Expression engine is the engine used in Sipelia™.

ring group

A ring group is a group of SIP entities that has its own unique SIP phone extension. All entities (or members) within a ring group are part of a call list, and all members get called when the ring group extension is called. The members of a ring group can either be called all at the same time, or successively at a set interval. The call stops ringing when any one of the members within a call list answers the call.

role

A role is a software component that performs a specific job within Security Center or Security Center SaaS.

Security Center

Security Center is a truly unified platform that blends IP video surveillance, access control, automatic license plate recognition, intrusion detection, and communications within one intuitive and modular solution. By taking advantage of a unified approach to security, your organization becomes more efficient, makes better decisions, and responds to situations and threats with greater confidence.

Security Desk

Security Desk is the unified user interface of Security Center. It provides consistent operator flow across all of the Security Center main systems, Omnicast™, Synergis™, and AutoVu™. The unique task-based design of Security Desk lets operators efficiently control and monitor multiple security and public safety applications.

self-signed certificate

A self-signed certificate is an *identity certificate* that is signed by the same entity whose identity it certifies, as opposed to a *certificate authority (CA)*. Self-signed certificates are easy to make and do not cost money. However, they do not provide all of the security properties that certificates signed by a CA aim to provide.

SIP client

A SIP client is a program with softphone functionality that users can install on their computers or mobile devices to make and receive voice or video calls from other SIP clients. A SIP client requires a SIP account, and typically includes a user interface from which users can manage calls and also view call-related video streams, if such a feature is supported. Examples of SIP clients are SIP phones, softphones, and SIP intercoms. Once Sipelia™ Client is installed and configured, Security Desk is also an example of a SIP client.

Sipelia™

Sipelia™ is a core module of Security Center. Using this module, Security Center users can make, receive, and manage SIP-based voice and video calls over a network. Running on the open source Session Initiation Protocol (SIP), Sipelia also integrates existing video and access control platforms with intercom systems, and allows users to log call activities.

Sipelia™ Server

Sipelia™ Server is the SIP server component of Sipelia. It receives and administers information about the different SIP endpoints, and essentially facilitates the communication between two or more endpoints that are communicating in a SIP environment. Sipelia Server also collates and stores important data, such as contact list information, SIP server settings, and call session recordings. Sipelia Server must be run by a Security Center plugin role, and therefore, must be installed on every Security Center server on which you intend to host the plugin role.

SIP endpoint

A SIP endpoint is the device or system that is at each end of a SIP call session. Examples of endpoints are hard-wired phones, voice mail systems, and intercoms. A SIP client such as a softphone, is another example

of an endpoint. Once Sipelia™ Client is installed and configured, Security Desk is considered both an endpoint and a SIP client.

SIP entity

A SIP entity is a Security Center entity that has SIP-related capabilities. In Security Center, examples of SIP entities are users, ring groups, and SIP devices such as SIP intercoms.

SIP extension

A SIP extension is a numeric value assigned to a SIP device so that the device can make and receive SIP calls. Typically, a SIP extension is also used to register the SIP device (to which it is assigned) to a SIP server. To be able to communicate with other SIP endpoints, every SIP entity (user, ring group, or intercom) in Security Center must have a unique SIP extension assigned to it.

SIP intercom

A SIP intercom is an intelligent SIP endpoint that provides two-way phone connectivity in a SIP environment. In Security Center, a SIP intercom is one of the established SIP entities, and it is the only SIP entity that is an actual device. The other SIP entities are Security Center users and ring groups.

SIP trunk

A SIP trunk is a SIP server that allows users to connect their existing SIP servers to other servers, thus extending their VoIP capabilities and allowing them to migrate their old PBX systems to a unified VoIP system. With an integrated dial plan, SIP trunks make it possible for SIP extensions that reside on different SIP servers to communicate with one another.

Transport Layer Security

Transport Layer Security (TLS) is a protocol that provides communications privacy and data integrity between two applications communicating over a network. When a server and client communicate, TLS ensures that no third party may eavesdrop or tamper with any message. TLS is the successor to the Secure Sockets Layer (SSL).

TURN server

The Traversal Using Relays around NAT (TURN) server enables communication between the Sipelia™ Mobile role and the mobile devices that are behind network address translators (NATs) or firewalls.

Unified report

The *Unified report* is an investigation task that you can use to query the *record types* available to you.

VoIP inventory

The *VoIP inventory* task is a maintenance task that reports on the SIP entities and all of the Security Center users with SIP accounts in your system.

Where to find product information

You can find our product documentation in the following locations:

- **Genetec™ TechDoc Hub:** The latest documentation is available on the TechDoc Hub.

  Can't find what you are looking for? Contact documentation@genetec.com.
- **Installation package:** The Installation Guide and Release Notes are available in the Documentation folder of the installation package. These documents also have a direct download link to the latest version of the document.
- **Help:** Security Center client and web-based applications include help, which explains how the product works and provide instructions on how to use the product features. To access the help, click **Help**, press F1, or tap the **?** (question mark) in the different client applications.

Technical support

Genetec™ Technical Assistance Center (GTAC) is committed to providing its worldwide clientele with the best technical support services available. As a customer of Genetec Inc., you have access to TechDoc Hub, where you can find information and search for answers to your product questions.

- **Genetec TechDoc Hub:** Find articles, manuals, and videos that answer your questions or help you solve technical issues.

  Before contacting GTAC or opening a support case, it is recommended to search TechDoc Hub for potential fixes, workarounds, or known issues.

  To access the TechDoc Hub, log on to Genetec Portal and click TechDoc Hub. Can't find what you're looking for? Contact documentation@genetec.com.
- **Genetec Technical Assistance Center (GTAC):** Contacting GTAC is described in the Genetec Advantage Description.

Technical training

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